

BEYOND THE BASE: FREQUENCY ALLOCATION IN ROPE

Anonymous authors

Paper under double-blind review

ABSTRACT

Rotary position embeddings (RoPE) typically tie their frequency placement to a single base. We study frequency allocation as a way to improve model quality within a chosen context range. Controlled experiments show that redistributing interior frequencies improves model performance even when the frequency range is fixed. Full sine–cosine geometry characterizes positional overlap and exposes allocation rankings missed by cosine-only proxies, while learned-coordinate effects guide how a pretrained model’s table can be modified. We introduce TailSpline, an explicit allocation that smooths the transition into the extended low-frequency tail. It retains the standard rotary operator and requires neither weight updates nor calibration. TailSpline substantially outperforms MrRoPE-Pro on RULER at both intermediate and target extension lengths, with a small observed native-window trade-off. Supporting experiments with Cosh, a different frequency transport for extrapolation, establish that useful allocation extends beyond this construction. Together, the analysis and experiments make internal frequency placement a practical dimension of RoPE design.

1 INTRODUCTION

A context window is useful when a model can use the information within it. For a model trained to length L , an extension factor s specifies a target length sL ; quality at the lengths served by that window determines its value. RoPE’s base selects a frequency range, while the placement of frequencies inside it determines the positional basis. We study how this internal allocation can improve model quality within the intended context range.

MrRoPE unifies frequency rescaling through mixed-radix conversion and proposes progressive extension (Tian et al., 2026); YaRN and LeRoPE further demonstrate the value of scaling and learning frequencies (Peng et al., 2024; Karypis et al., 2026). We investigate the positional structure and quality of these allocations. Paired experiments hold the frequency endpoints fixed and show that interior placement improves both learned extrapolation and frozen-model performance. Additional controls distinguish its effects from total frequency displacement and learned coordinate assignment.

Full sine–cosine geometry reveals how distinct frequencies can supply overlapping positional directions. For frozen extension, a model already uses its native frequencies; we introduce TailSpline to allocate extension through a smooth connection to the low-frequency tail. A single static table improves clean RULER over MrRoPE-Pro at both $2L$ and $4L$, by 3.39 and 11.72 percentage points, with a small observed native-window trade-off. Cosh provides a complementary frequency transport for extrapolation, used in paired training to establish the wider value of changing z .

Our contributions are:

- **Identify the value of allocation.** Controlled training and frozen interventions separate interior placement from range, total displacement, and learned coordinate assignment.
- **Characterize its positional structure.** Full-pair geometry quantifies finite-window overlap; a discrete kernel criterion distinguishes spectral changes from fixed changes of basis.
- **Improve quality within the extended window.** TailSpline gives a closed-form frozen allocation with strong paired task gains at intermediate and target lengths. Cosh supplies a supporting extrapolation instance.

2 ALLOCATION AS A DESIGN VARIABLE

For $K = d_{\text{rot}}/2$ pairs, standard RoPE uses $\omega_k = b^{-k/K}$. Any strictly ordered positive table admits

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (1)$$

The endpoints (a, R) fix the sampled log-frequency interval, called the support; $K - 2$ interior coordinates define allocation. Every geometric table has $z_k = k/(K - 1)$ irrespective of base. The full space $\mathcal{Z}_K = \{0 = z_0 < \dots < z_{K-1} = 1\}$ includes analytic subfamilies and positive-gap learned parameterizations.

Allocation applies to a chosen frequency range, whether that range is retained or extended. For deployment relative to a native table, write

$$\omega'_k = \omega_k^N s^{-m_k}, \quad d_k = \log(\omega_k^N / \omega'_k) = m_k \log s. \quad (2)$$

Changing frequencies preserves RoPE’s relative-position composition and norm preservation (Su et al., 2024). Multiplying cosine/sine amplitudes by g separately scales the rotary Q/K contribution by g^2 . Fixed-support experiments isolate z ; complete methods may jointly choose allocation, range and amplitude.

Geo denotes each protocol’s geometric reference and Native a released model’s inherited table. Standard and midpoint geometric grids share z but differ by an overall frequency offset. Exact grid and endpoint conventions are in Appendix D.2.

What each control fixes. With fast endpoint unchanged and slow endpoint divided by s , $z'_k = (Rz_k + \log s m_k)/(R + \log s)$. For a transition with n gaps, let $\epsilon_q = m_q - m_{q-1}$, $m_0 = 0$, $m_n = 1$. The controls fix different quantities:

Held fixed	Meaning	Remaining intervention
Actual endpoints	Sampled support (a, R)	Interior allocation
$\sum_q \epsilon_q = 1$	Added log span $\log s$	Total displacement and shape
$D = \log s \sum_k m_k$	Total log displacement (dose)	Residual shape
Frequency multiset	Same supplied spectrum	Coordinate assignment
Gain g	Rotary score amplitude	Frequency effects

At fixed band and $s > 1$, $\sum_{q=1}^n q\epsilon_q = n - \sum_{q=1}^{n-1} m_q$: fixing dose already fixes the increment centroid. It is not a second independent control.

3 FREQUENCY ALLOCATION MATTERS AT FIXED RANGE

We first isolate the effect of interior placement by fixing the frequency endpoints. We then examine how that effect depends on training, total frequency displacement, and the model’s learned use of its coordinates.

3.1 A PAIRED INTERVENTION AT FIXED SUPPORT

Reallocating interior frequencies improves learned extrapolation at the same frequency endpoints. We train paired 151.9M models with $K = 32$ rotary pairs at length 256.

Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The matched geometric control (Geo) uses the training-base configuration $b = L_{\text{train}}$ from FMRoPE (Oka et al., 2026); the EVQ-Cosh intervention anchors its quantiles with $\tau = 4$ to the same highest and lowest frequencies and log-span (§5.2). Only the 30 interior z coordinates differ.

With the training support retained at evaluation, mean Cosh-minus-Geo tail NLL is +0.026 at the training length and −0.281/ −0.176/ −0.146 at 512/1024/2048 tokens. All three paired seeds improve at every extended length (Fig. 1b).

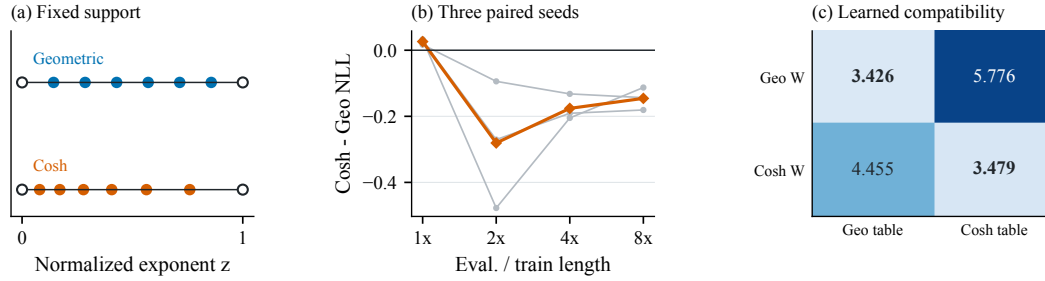


Figure 1: **Fixed-support allocation and learned compatibility.** (a) Fixed endpoints leave internal spacing free ($K = 8$ illustration). (b) Three paired training seeds favor Cosh at extended lengths with retained support; native NLL increases. (c) Two-seed mean tail NLL for the 151.9M weights-by-derived-table crossing at 1K. Rows freeze trained weights; columns install runtime tables. The matched diagonals outperform the crossed installations.

Retargeting both supports to $b = L_{\text{eval}}$ while retaining each trained checkpoint and normalized allocation reverses the ordering: Cosh-minus-Geo NLL becomes $+0.060 / +0.227 / +0.460$, with all three seeds favoring Geo (Fig. 10a). Allocation is a useful design variable, and its preferred setting depends on the support policy.

The effect extends beyond one allocation family. A 50.9M short-training factorial (8.39M tokens per arm) with three paired seeds tests two bases, two training lengths and three head dimensions. Reference Cosh, its preassigned $1.25\times$ strength, and a displacement-matched exponential improve OOD NLL in 7/12, 10/12, and 9/12 configurations. Cosh and the matched exponential perform similarly (mean difference $+0.00074$ NLL, interval $[-0.006, 0.008]$). These results support nonuniform allocation across families, with strength-dependent trade-offs; full settings and a separate standard-grid control are in Appendices B.3 and C.1.

3.2 THE EFFECT REMAINS IN FROZEN MODELS

On OLMo-2-1B and Qwen-2.5-1.5B, we fix the checkpoint, sampled endpoints, 64 pairs, attention amplitude, input rows, and decoder. The range extends from the native fast frequency to one-quarter of the native slow frequency. On OLMo’s nine held-out RULER tasks (Hsieh et al., 2024), uniform normalized exponents score 0.56%, a coarse ramp 61.04%, and a residual-guided profile 60.47%. Qwen’s four-task development comparison gives 57.75/64.00/66.50% (Appendix F.2). Both reallocations yield large OLMo gains while retaining the same sampled range.

Internal shape beyond total displacement. Fixed endpoints do not fix the sum of native-relative shifts $\sum_k d_k$. A separate frozen OLMo comparison holds that sum, endpoints, band and gain fixed: the symmetric profile BM and uniform transition MrRoPE-Uni score 51.32 and 32.12% in a six-task, 48-input 16K follow-up comparison. Total displacement therefore does not explain this score difference (Appendix G.8).

A learned model uses more than a frequency range. Crossing the runtime tables of the paired 151.9M models tests whether weights can use a different allocation interchangeably. Both seeds favor the table derived from their training allocation (Fig. 1c; Appendix B.2). A complementary frozen intervention permutes the same frequency multiset across coordinates and changes model performance (Appendix I.4). These findings connect the two operating settings: learning can adapt weights to an allocation, whereas frozen extension must work with the model’s existing frequency-coordinate association.

4 HOW ALLOCATION CHANGES POSITIONAL STRUCTURE

The controlled gains raise a structural question: what changes when the frequency range stays fixed? We examine the positional directions supplied by complete rotary pairs. Their overlap connects

frequency spacing to redundancy within a finite window; the model learns how to combine these directions with content.

4.1 COMPLETE ROTARY PAIRS AND POSITIONAL OVERLAP

For a fixed content pair, one rotary channel contributes $C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to an attention logit, where Δ is relative token separation. Both coefficients depend on content; the positional object is

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

To avoid privileging a content phase, let $x_\omega(\Delta) = [\cos(\omega\Delta), \sin(\omega\Delta)]$ and define $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$, $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ under a declared separation measure. For nonsingular self-Grams, the singular values of $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ are the canonical correlations, or cosines of principal angles, of the two spaces (Björck & Golub, 1973). Their mean square,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

measures shared positional directions and is invariant to invertible basis changes within either pair. Let Γ be the Gram matrix of the concatenated block-whitened pairs, with diagonal blocks I_2 . The average pairwise overlap $\bar{c}$ gives the exact Rényi-2 effective-rank identity

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

Here $\text{tr}(\Gamma^2)$ is the frame potential of the normalized positional functions (Benedetto & Fickus, 2003). The identity specializes this classical measure to full rotary pairs: increasing pairwise overlap concentrates the same $2K$ coordinates into fewer effective positional directions.

4.2 FINITE-WINDOW CONCENTRATION OF SLOW DIRECTIONS

Proposition 1 (Shared slow-frequency subspace). *Under uniform measure on $[0, L]$, with $t = \Delta/L$, as $x = \omega L \rightarrow 0$, $\cos(xt) \rightarrow 1$ and $\sin(xt)/x \rightarrow t$, so V_ω converges to $\text{span}\{1, \Delta\}$. For $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$, squared chordal separation satisfies $2 - \|Q_{\omega\nu}\|_F^2 = \mathcal{O}(\epsilon^4)$.*

Distinct slow frequencies therefore approach a shared two-dimensional positional space. On the standard grid $b = 500,000$, $K = 64$, $L = 4096$, the 23 pairs with $\omega L \leq 1$ occupy 46 coordinates but have block-whitened effective rank $r_2 = 2.00013$. At $b = 256$, $K = 32$, $L = 256$, the eight slowest pairs span $\omega L \in [1.19, 4]$ and have effective rank $r_2 = 2.11$ (Appendix A.2). Their overlap is substantial even though every frequency has $\omega L > 1$.

Why use complete rotary pairs? A cosine-only overlap proxy can reverse the allocation ordering: explicit four-pair tables with identical endpoints give $C_{\cos}(A) = 0 < C_{\cos}(B)$, yet $r_2(A) = 6.29 < 8.00 \approx r_2(B)$ because sine-cosine cross terms also carry positional overlap (Appendix A.4). The complete-pair measure captures this structure independently of phase.

Design implication. The same number of rotary pairs can supply very different positional diversity. Reallocating frequencies changes this overlap, providing a concrete reason to consider their spacing during learning. Longer-range coverage also matters: a full-rank parity lattice can repeat outside its reference window (Appendix A.3). We therefore design allocations for a stated operating setting, balancing finite-window resolution with frequency coverage.

4.3 CHANGING FREQUENCIES CHANGES THE POSITIONAL KERNEL

The learned-compatibility experiments raise a second question: can a fixed change of Q/K coordinates absorb a frequency change? For the exact bilinear rotary kernel, the rotation spectra give a simple criterion.

Corollary 2 (Integer-position equivalence). *Let Ω, Λ contain K frequencies in $(0, \pi)$ and $\mathcal{R}_\Omega(d) = \text{diag}_k R(\omega_k d)$. Position-independent invertible A, B satisfy $A^\top \mathcal{R}_\Omega(d) B = \mathcal{R}_\Lambda(d)$ for $d = 0, 1$ if and only if the frequency multisets agree.*

At $d = 0$, $B = (A^\top)^{-1}$; at $d = 1$, similarity preserves $\{e^{\pm i\omega_k}\}$. In $(0, \pi)$ this identifies the multiset. A block permutation proves the converse for every integer d . This spectral criterion separates changing the supplied frequencies from reassigning an unchanged spectrum, complementing STRING’s structural characterization (Schenck et al., 2025). The assumptions concern exact kernels for all content vectors; the full proof and aliasing cases are in Appendix A.6.

5 EXPLICIT FREQUENCY ALLOCATIONS

Allocation changes how a fixed number of rotary pairs covers the frequency range. Our main frozen construction, TailSpline, designs this change around the model’s native table. Cosh supplies a complementary transport used to study extrapolation through paired training.

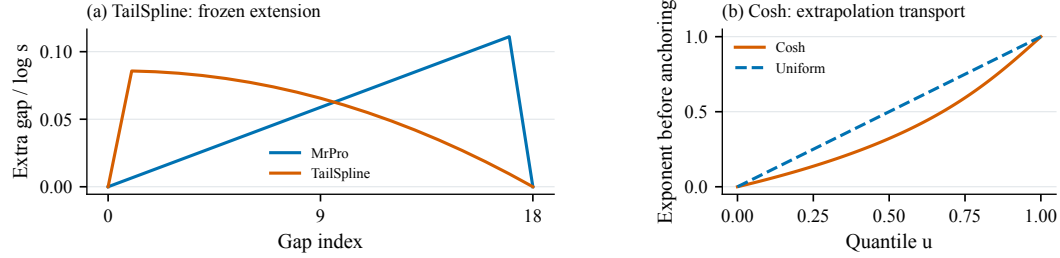


Figure 2: **Explicit ways to change allocation.** (a) TailSpline allocates extra log-frequency span around a native table, smoothing the connection to the extended low-frequency tail ($n = 17$). (b) Cosh supplies a complementary extrapolation transport, shown as quantiles before endpoint anchoring ($\tau = 2$).

5.1 TAILSPLINE: ALLOCATING EXTENSION AROUND A NATIVE TABLE

A frozen model already uses its native frequencies. We retain its high-frequency band and divide its low-frequency tail by s , while designing the intervening allocation. Following the 32/1-turn boundaries used by MrRoPE (Tian et al., 2026), let l be the last pair making more than 32 turns over the native window and h the first making less than one. For a nonempty transition, put $n = h - l \geq 1$ and $q = \text{clip}(k - l, 0, n)$. The transition accumulates increments $\epsilon_q = m_q - m_{q-1}$, with $m_0 = 0$, $m_n = 1$.

The increments have a direct spectral meaning. For a geometric native grid with gap $c = \log b/K$,

$$x'_{l+q} - x'_{l+q-1} = c + \epsilon_q \log s. \quad (6)$$

Their unit sum allocates exactly $\log s$ of extra span across the band. Beyond it, constant displacement $m = 1$ preserves native adjacent gaps: the extra gap returns to zero. The remaining choice is how to reach this junction.

MrRoPE-Pro uses increasing increments, $m_q = q(q+1)/[n(n+1)]$. We instead smooth the transition into the fully interpolated tail, whose subsequent increments are zero:

$$\min_{\epsilon: \sum_{q=1}^n \epsilon_q = 1} \sum_{q=1}^{n-1} (\epsilon_{q+1} - \epsilon_q)^2 + \epsilon_n^2. \quad (7)$$

The first term smooths neighboring extra gaps; the second matches the low-frequency junction. Leaving the high-frequency entry unpenalized prioritizes the tail connection while preserving the high-frequency outer band. The fully interpolated tail supplies the zero-extra-gap boundary used to close the transition.

Theorem 3 (Finite-grid TailSpline). *The unique minimizer of Eq. (7) has positive increments and cumulative profile*

$$\epsilon_q = \frac{3(n+q)(n-q+1)}{n(n+1)(2n+1)}, \quad m_q^{\text{TS}} = \frac{q(3n^2 + 3n + 1 - q^2)}{n(n+1)(2n+1)}. \quad (8)$$

Strict convexity and the tridiagonal stationarity system give Eq. (8) (Appendix H). Adding the entry penalty ϵ_1^2 yields the symmetric construction BM, with increments proportional to $q(n - q + 1)$ (Appendix G.3). Relative to MrPro, TailSpline accepts a larger entry jump and reduces the terminal jump by the exact factor $3/(2n + 1)$: 0.086 for Llama’s $n = 17$ and 0.081 for OLMo’s $n = 18$ (Fig. 2a). These guarantees solve the declared objectives; task value is tested below.

What the comparison isolates. TailSpline and MrPro retain the same outer bands and endpoints while changing both total displacement and transition shape. An additional control C matches TailSpline’s total displacement to test the remaining shape difference (Appendix H.2). The completed classic diagnostic gives T-C -0.41 points in Full-13 AUC, with interval $[-2.63, 1.82]$; T uses batch 1 and C batch 2, leaving this finer ordering unresolved (Appendix H.8).

Installation. (1) Read the public native grid, reference length and scale s ; (2) compute the 32/1-turn band; (3) evaluate $m_k, \omega'_k = \omega_k^N s^{-m_k}$ and $g = 1 + 0.1 \ln s$; (4) install before prefill, retaining one table across layers and input lengths. The baseline shares g . Construction costs $O(K)$ and leaves the rotary computation unchanged. Neither constructor reads weights, activations or calibration outputs.

5.2 COSH AS A SUPPORTING EXTRAPOLATION TRANSPORT

Cosh changes z through a monotone map of normalized exponents. Motivated by finite-window overlap among slow pairs, it balances frequency concentration against slow-end tail mass. The density objective and its unique minimizer are derived in Appendix A. Its inverse CDF is

$$Q_\tau(u) = 1 - \frac{\operatorname{arcsinh}((1-u) \sinh \tau)}{\tau}, \quad u_k = \frac{k + 1/2}{K}. \quad (9)$$

The uniform-density limit is recovered as $\tau \rightarrow 0$. Affine anchoring of $Q_\tau(u_k)$ to the chosen endpoints gives a convex allocation that moves interior frequencies toward the faster end; Appendix D.2 gives the installed grid and Eq. (52). This is the transport used in the fixed-range training intervention of §3.1. The strength τ determines the amount moved: the fixed-support reference is $d_{\text{head}}/\sqrt{L_{\text{train}}}$, and the MLA extrapolation recipe uses 1.414. The factorial compares neighboring strengths and a matched exponential transport (Appendix B.3). Every installation retains the standard rotary operator.

6 MODEL QUALITY WITHIN THE TARGET WINDOW

The main task evaluation measures quality within the intended window under one static TailSpline table. Supporting Cosh experiments examine another way to transport frequencies for extrapolation. Table 1 summarizes the distinct interventions.

Table 1: **Controls and operating settings.** The paired training interventions identify allocation; subsequent evaluations test the two constructions in use. Fixed support means actual sampled endpoints.

Protocol	Weight updates	Support fixed?	Main question
151.9M, three seeds	Paired training	Yes	Interior allocation
432M MLA, three seeds	500M tokens	No (midpoints)	Learning utility
TailSpline–MrPro	None	Yes, extended	Frozen deployment
TailSpline–C (E1)	None	Yes; dose too	Shape diagnostic
Frequency-slot swap	None	Same multiset	Learned assignment
Native comparison	None	Different tables	Original-window cost

6.1 LONG-CONTEXT GAINS FROM A FROZEN TAILSPLINE TABLE

Frozen Llama-3-8B-Instruct uses $s = 4$, native window 8192, band $[18, 35]$ and gain $1 + 0.1 \ln 4$. TailSpline and MrPro share weights, outer frequencies, endpoints, gain, inputs and decoder; their interior allocation and resulting total displacement differ. Each arm installs one static table before

prefill. This evaluates quality within a fixed target range. MrRoPE’s published Llama comparison uses $s = 16$ (8K to 128K), while its Qwen comparison uses $s = 4$ (32K to 128K) (Tian et al., 2026); our paired Llama comparison uses $s = 4$.

Clean RULER-200 at 32K. The confirmation panel uses 200 source-order inputs per RULER task (2,600 paired prompts), without depth selection or content padding. Actual lengths are 28,270–32,606 tokens; both arms use unpadded batch 1 and the same official scorer. TailSpline scores 68.27% versus 56.54%, a task-equal difference of +11.72 points (paired 95% interval [10.32, 13.11]). It wins 12 of 13 task means. Retrieval, tracking, aggregation and QA all have positive family contrasts; leave-one-task-out differences remain positive (Appendix H.7). The primary endpoint is the task-equal score at 32K.

Quality at the intermediate length. The completed clean 16K panel uses 50 inputs per task (650 pairs). TailSpline scores 86.09% versus 82.71%, a +3.39-point Full-13 gain with interval [1.53, 5.34]. Retrieval improves by 4.50 points; the RULER QA family scores 68/60%, a +8-point gain with interval [1, 15]. Together with the +9-point QA gain at 32K, these comparisons show task benefits at both tested extended lengths (Table 2). The complete task breakdown is in Appendix H.10.

Table 2: **Task gains at both tested extended lengths.** Clean RULER Full-13 task-equal scores; paired intervals condition on each length’s inputs.

Length	Inputs/task	T (%)	P (%)	T-P (pp)	95% interval
16K (2L)	50	86.09	82.71	+3.39	[1.53, 5.34]
32K (4L)	200	68.27	56.54	+11.72	[10.32, 13.11]

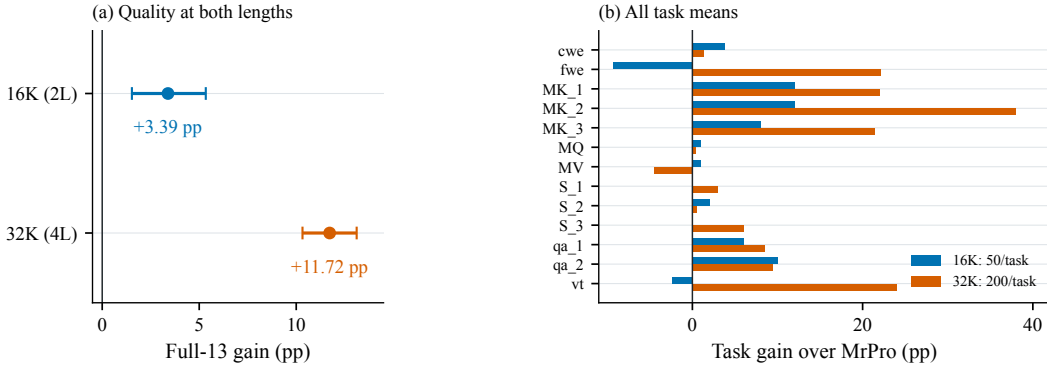


Figure 3: **TailSpline improves task quality at both 2L and 4L.** Clean source-order RULER with one static $s = 4$ table. (a) Full-13 task-equal gains and paired 95% intervals. (b) All task differences at each length; 16K uses 50 inputs per task and 32K uses 200. MK, MQ, MV and S denote multikey, multiquery, multivalue and single-key retrieval.

Native-window trade-off. TailSpline’s +11.72-point clean long-context gain accompanies a small native-window trade-off. On the matched 130-prompt 8K task reference, it scores 89.74% versus original Native’s 91.88%: an observed 2.14-point difference, or 2.33% relative reduction (paired T-Native interval [−6.14, 1.92] points). This native estimate uses the classic padded panel; the long-context gain uses the clean source-order panel. Whole-prefix 8K PPL is 5.29528 versus 5.27592, a 0.37% relative increase. The same static table serves both windows, without length-specific switching. Paired intervals and the separate equal-dose T/C diagnostic are reported in Appendix H.8.

Classic length curves and transfer. The classic Llama panel has 10 paired inputs per task at 8/16/32K (390/arm), with output-independent depth selection and prefix padding. PPL uses 32 Proof-Pile and 14 PG-19 test documents (138 LM rows/arm) (Azerbayev et al., 2022; Rae et al., 2019). Full-13 AUC rises from 75.60 to 78.80% (+3.20 points, 95% interval [0.65, 5.79]); its nested NIAH

subset improves by 3.75 points. PPL AUC changes by -0.097% , with small 8/16K increases; QA-family AUC changes by -1.25 points. On OLMo-2-1B-Instruct, the same formula at band [14, 32] and 4/8/16K raises Full-13 AUC from 17.36 to 66.60% (+49.23 points, interval [45.34, 53.04]), with all task AUC differences positive. Full curves (Fig. 15), aggregation weights and termination counts are in Appendix H.

TailSpline on natural QA. The frozen five-task Natural-QA pool contains 631 paired questions from 524 source-context clusters, retokenized for Llama with actual inputs 3694–16304 tokens. Both methods use the same $s = 4$ tables, unpadded batch 1, prompts and generation caps. Whole-response token F1 is averaged within each task, then equally across tasks; uncertainty resamples source-context clusters. Overall F1 is 41.08/40.88% for T/P, a $+0.20$ -point difference with interval $[-1.53, 1.89]$. Appendix H.9 reports the complete pool and native-length strata. The observed F1 scores are close, with the interval spanning both orderings. T/P end with EOS on 625/624 inputs, reach the cap on 6/7, and produce no empty responses. Full task scores and protocol are in Appendix H.9.

6.2 SUPPORTING EXTRAPOLATION EVIDENCE WITH COSH

A 432M MLA-style model uses 16 rotary pairs, 500M training tokens at 8K and three seeds (DeepSeek-AI, 2024a;b), with fixed Cosh strength $\tau = 1.414$. All arms use the same FineWeb-Edu evaluation chunks. This learning recipe uses unanchored midpoint tables; Table 1 distinguishes it from the fixed-endpoint intervention. Cosh improves all five extended lengths in every seed; 16K PPL falls from 138.8 to 95.6. The benefit appears at half and three-quarters of training and persists under a common wavelength blend (Fig. 4; Appendix E.1).

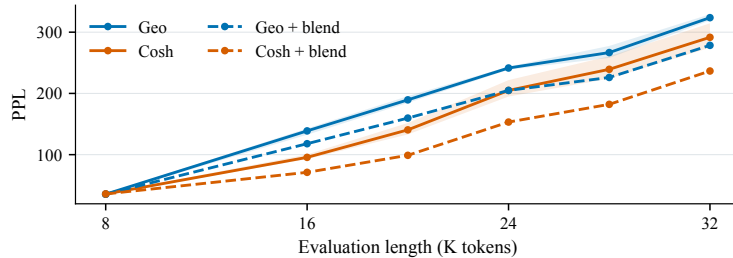


Figure 4: **Cosh improves extrapolation across training seeds.** Three-seed mean and range for the 432M MLA model; dashed curves use the common wavelength blend ($s = 4$). Evaluation uses common text chunks across arms (Appendix E.1).

A 750M single-seed continuation also improves 16K PPL and 8K answer-token exact match, with a small 4K cost. Matched OLMo adaptation uses explicit long-phase exposure and exhibits native/extended-length trade-offs. Their full budgets, scores and supervision remain in Appendix E.3.

7 RELATED WORK

Frequency scaling and frozen deployment. RoPE (Su et al., 2024), base analyses and FMRoPE connect frequency range with extrapolation (Liu et al., 2024; Xu et al., 2024; Oka et al., 2026). Position interpolation, NTK scaling, YaRN and MrRoPE provide established rescaling policies (Chen et al., 2023a; bloc97, 2023; Peng et al., 2024; Tian et al., 2026). MrRoPE’s mixed-radix increments are our nearest frozen comparator; TailSpline retains its outer bands and changes the transition objective. LongRoPE and LongRoPE2 combine search and learning, the latter using mixed-context training (Ding et al., 2024; Shang et al., 2025). Their deployment results have different training budgets from a frozen table swap.

Frequency function and learning. Round and Round studies positional and semantic frequency use; its p -RoPE gains include Gemma training from scratch (Barbero et al., 2025). HoPE studies high-frequency positional encoding (Chen et al., 2025); Massive Values studies large Q/K components (Jin et al., 2025). FoPE changes Fourier components, while LeRoPE learns frequencies and derives their attention/value gradients (Hua et al., 2025; Karypis et al., 2026). We build on frequency design by

isolating the value of interior placement and connecting positional structure to explicit constructions for learning and frozen extension.

Group structure and changes of basis. STRING characterizes differentiable matrix encodings satisfying an identity and a group-like relative-translation law, and relates them to RoPE through an orthogonal basis (Schenck et al., 2025). GRAPE connects group generators, learned frequencies and frequency–norm coupling (Zhang et al., 2026). Our integer-position corollary uses rotation spectra to separate a change in frequencies from a fixed change of basis.

Task structure and positional dependence. PINE changes inter-document masks and ordering to target document-level invariance (Wang et al., 2024b); PRoPE encodes relative camera geometry (Li et al., 2025). Selective RoPE and RePo learn input-dependent rotations or positions (Movahedi et al., 2026; Li et al., 2026). These modify different objects from a static frequency allocation. HELMET separates application categories whose behavior synthetic retrieval alone does not predict (Yen et al., 2024), motivating distinct retrieval, natural-QA and native-window evidence rather than a single pooled score.

Our contribution connects three additions: interventions that hold actual frequency endpoints fixed, complete-pair examples that expose phase-dependent proxy rankings, and a closed-form tail transition validated at two clean extension lengths. The fixed-range controls establish the design freedom; the geometric analysis characterizes what it changes; the TailSpline comparison measures its practical effect.

8 DISCUSSION AND CONCLUSION

A frequency range leaves a consequential choice open: where to place the frequencies within it. Our controlled interventions establish that this choice can improve learned extrapolation and frozen-model task performance. Full-pair geometry characterizes the resulting positional overlap, while learned-coordinate effects clarify how frequency changes interact with a pretrained model.

TailSpline provides an explicit native-relative allocation that retains standard rotary computation. Cosh offers a complementary transport for extrapolation and supports the broader value of changing z . The clean RULER comparisons show TailSpline gains over MrPro at both $2L$ and $4L$, with a small observed native-window trade-off. Natural-QA evaluation gives close observed F1 scores for TailSpline and MrPro. Together, these results establish frequency allocation as a practical design dimension for quality within a chosen context window.

ETHICS STATEMENT

This work studies frequency allocation in existing language models using public research corpora and evaluation suites under their stated licenses. It introduces no human-subject study or deployed decision system. The proposed constructions retain the rotary operator and add no trainable parameters; adaptation experiments use the weight updates specified in their protocols.

REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. The source archive provides the active TeX, vector figures, numerical summaries and CPU checks of the Gram-matrix and allocation identities. It also includes the frozen two-arm data-preparation, table-construction, generation, scoring and paired-report entrypoints with their local dependencies. The runtime guide specifies commands, public data sources and the pinned RULER revision.

The core fixed-support identification uses six runs of 499,974,144 tokens each; the 432M comparison uses six 500M-token runs, and the 750M continuation uses two 500M-token runs. These are protocol-specific training budgets, with the corresponding recipes and additional studies detailed in the appendices.

AI USE STATEMENT

We used generative AI tools to assist with mathematical discussion, code development, literature organization, and manuscript editing. Experimental results are drawn from recorded model runs and numerical analyses. The authors are responsible for the correctness and originality of the submitted text, mathematical arguments, experimental claims, and accompanying code.

REFERENCES

- Zhangir Azerbayev, Edward Ayers, and Bartosz Piotrowski. Proof-Pile. Dataset, 2022. URL <https://huggingface.co/datasets/hoskinson-center/proof-pile>.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3119–3137, 2024. doi: 10.18653/v1/2024.acl-long.172. URL <https://aclanthology.org/2024.acl-long.172/>.
- Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e6d58fc68c0f3c36ae6e0e64478a69c0-Abstract-Conference.html.
- John J. Benedetto and Matthew Fickus. Finite normalized tight frames. *Advances in Computational Mathematics*, 18:357–385, 2003. doi: 10.1023/A:1021323312367.
- Åke Björck and Gene H. Golub. Numerical methods for computing angles between linear subspaces. *Mathematics of Computation*, 27(123):579–594, 1973. doi: 10.1090/S0025-5718-1973-0348991-3.
- Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, Usvsn Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*, pp. 95–136, 2022. doi: 10.18653/v1/2022.bigscience-1.9. URL <https://aclanthology.org/2022.bigscience-1.9/>.
- bloc97. NTK-aware scaled RoPE allows LLaMA models to have extended (8k+) context size without any fine-tuning and minimal perplexity degradation. Reddit, r/LocalLLaMA, 2023. URL https://www.reddit.com/r/LocalLLaMA/comments/141z7j5/ntkaware_scaled_rope_allows_llama_models_to_have/.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023a. doi: 10.48550/arXiv.2306.15595. URL <https://arxiv.org/abs/2306.15595>.
- Yuhan Chen, Ang Lv, Jian Luan, Bin Wang, and Wei Liu. HoPE: A novel positional encoding without long-term decay for enhanced context awareness and extrapolation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 23044–23056, 2025. doi: 10.18653/v1/2025.acl-long.1123. URL <https://aclanthology.org/2025.acl-long.1123/>.
- Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. LongLoRA: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307*, 2023b. doi: 10.48550/arXiv.2309.12307. URL <https://arxiv.org/abs/2309.12307>.
- DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024a. doi: 10.48550/arXiv.2405.04434. URL <https://arxiv.org/abs/2405.04434>.
- DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024b.
- Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of PMLR, pp. 11091–11104, 2024. URL <https://proceedings.mlr.press/v235/ding24i.html>.

- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. doi: 10.48550/arXiv.2407.21783. URL <https://arxiv.org/abs/2407.21783>.
- Robert M. Gray and David L. Neuhoff. Quantization. *IEEE Transactions on Information Theory*, 44(6):2325–2383, 1998. doi: 10.1109/18.720541.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020. doi: 10.18653/v1/2020.coling-main.580. URL <https://aclanthology.org/2020.coling-main.580/>.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=kIoBbc76Sy>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Ermo Hua, Che Jiang, Xingtai Lv, Kaiyan Zhang, Youbang Sun, Yuchen Fan, Xuekai Zhu, Biqing Qi, Ning Ding, and Bowen Zhou. Fourier position embedding: Enhancing attention’s periodic extension for length generalization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 24932–24949, 2025. URL <https://proceedings.mlr.press/v267/hua25b.html>.
- Mingyu Jin, Kai Mei, Wujiang Xu, Mingjie Sun, Ruixiang Tang, Mengnan Du, Zirui Liu, and Yongfeng Zhang. Massive values in self-attention modules are the key to contextual knowledge understanding. *arXiv preprint arXiv:2502.01563*, 2025. URL <https://arxiv.org/abs/2502.01563>.
- Petros Karypis, Sean O’Brien, Shreyas Kadekodi, Rui Zhu, and Julian McAuley. LeRoPE: Learnable RoPE frequencies improve language modeling. *arXiv preprint arXiv:2607.10134*, 2026. doi: 10.48550/arXiv.2607.10134. URL <https://arxiv.org/abs/2607.10134>.
- Huayang Li, Tianyu Zhao, Deng Cai, and Richard Sproat. RePo: Language models with context re-positioning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/64002>. arXiv:2512.14391.
- Ruilong Li, Brent Yi, Junchen Liu, Hang Gao, Yi Ma, and Angjoo Kanazawa. Cameras as relative positional encoding. *arXiv preprint arXiv:2507.10496*, 2025. URL <https://arxiv.org/abs/2507.10496>.
- Xiaoran Liu, Hang Yan, Chenxin An, Xipeng Qiu, and Dahua Lin. Scaling laws of RoPE-based extrapolation. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=J07k0SJ5V6>. arXiv:2310.05209.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017. URL <https://openreview.net/forum?id=Byj72udxe>. arXiv:1609.07843.
- Sajad Movahedi, Timur Carstensen, Arshia Afzal, Frank Hutter, Antonio Orvieto, and Volkan Cevher. Selective rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=AQo1SEElNb>.
- Yui Oka, Itsumi Saito, Kyosuke Nishida, and Kuniko Saito. Frequency bands in RoPE: Base frequency and context length shape the interpolation–extrapolation trade-off. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=PR1PPxvG9Q>.

- Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024. doi: 10.48550/arXiv.2406.17557. URL <https://arxiv.org/abs/2406.17557>.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=wHBfxhZulu>.
- Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. *arXiv preprint arXiv:1911.05507*, 2019. URL <https://arxiv.org/abs/1911.05507>.
- Connor Schenck, Isaac Reid, Mithun George Jacob, et al. Learning the ropes: Better 2d and 3d position encodings with string. *arXiv preprint arXiv:2502.02562*, 2025. URL <https://arxiv.org/abs/2502.02562>.
- Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen, and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 54203–54218, 2025. URL <https://proceedings.mlr.press/v267/shang25a.html>.
- Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pp. 843–852, 2015. URL <https://proceedings.mlr.press/v37/srivastava15.html>.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025. doi: 10.48550/arXiv.2501.00656. URL <https://arxiv.org/abs/2501.00656>.
- Qingyuan Tian, Wenhong Zhu, Xiaoran Liu, Xiaofeng Wang, and Rui Wang. MrRoPE: Mixed-radix rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/69413f87e5a34897cd010ca698097d0a-Abstract-Conference.html.
- Suyuchen Wang, Ivan Kobyzev, Peng Lu, Mehdi Rezagholizadeh, and Bang Liu. Resonance RoPE: Improving context length generalization of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 586–598, 2024a. doi: 10.18653/v1/2024.findings-acl.32. URL <https://aclanthology.org/2024.findings-acl.32/>.
- Ziqi Wang, Hanlin Zhang, Xiner Li, Kuan-Hao Huang, Chi Han, Shuiwang Ji, Sham M. Kakade, Hao Peng, and Heng Ji. Eliminating position bias of language models: A mechanistic approach. *arXiv preprint arXiv:2407.01100*, 2024b. URL <https://arxiv.org/abs/2407.01100>.
- Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Pan Zhang, Yuhang Cao, Jian Tong, Haodong Duan, Qipeng Guo, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. VideoRoPE: What makes for good video rotary position embedding? In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 66118–66136, 2025. URL <https://proceedings.mlr.press/v267/wei25h.html>.
- Mingyu Xu, Xin Men, Bingning Wang, Qingyu Zhang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. Base of RoPE bounds context length. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2773.

- URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/9f12dd32d552f3ad9eaa0e9dfec291be-Abstract-Conference.html.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. doi: 10.48550/arXiv.2412.15115. URL <https://arxiv.org/abs/2412.15115>.
- Howard Yen, Tianyu Gao, Minmin Hou, Ke Ding, Daniel Fleischer, Peter Izsak, Moshe Wasserblat, and Danqi Chen. Helmet: How to evaluate long-context language models effectively and thoroughly. *arXiv preprint arXiv:2410.02694*, 2024. URL <https://arxiv.org/abs/2410.02694>.
- Yifan Zhang, Zixiang Chen, Yifeng Liu, Zhen Qin, Huizhuo Yuan, Kangping Xu, Yang Yuan, Quanquan Gu, and Andrew Chi-Chih Yao. Group representational position encoding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/5cb58625f49ddf70fe2d527e9e4bbae5-Paper-Conference.pdf.
- Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 77539–77557, 2025. URL <https://proceedings.mlr.press/v267/zhao25m.html>.

GUIDE TO THE SUPPLEMENTARY MATERIAL

The supplementary material follows the argument from the independent value of z to positional structure, explicit constructions and task validation. Table 3 maps claims to their supporting material; Table 4 distinguishes the experimental contracts. A short evidence path is fixed-support identification (Appendix B.1), clean frozen confirmation (Appendix H.7), and natural QA (Appendix H.9). The remaining studies provide construction proofs, learning utility and historical controls with their own protocols.

Table 3: Where each part of the argument is developed.

Question	Supporting material
How is allocation value identified?	Fixed-support training (App. B.1); multi-shape factorial (App. B.3); frozen controls (App. F.2).
How are the constructions derived?	Full-pair geometry and Cosh (App. A); TailSpline (App. H); BM (App. G.3).
What are the complete learning and deployment results?	Continuation and MLA (Apps. D, E.1); adaptation and readout (App. F); frozen profiles and QA (App. G); TailSpline (App. H).
What controls delimit the conclusions?	Range, crossed-table and slot interventions (App. I); profile diagnostics (App. J.6); interval design (App. J); video and architecture (Apps. D.7, D.9).

Table 4: **Protocol glossary: which comparison supports which conclusion.** Results are not pooled across rows. Geometry and construction theorems characterize their declared objects; task value is measured by the corresponding model evaluation.

Protocol	Weight state	Table and matched control	Evidence role
Fixed-support identification	Paired scratch training or frozen intervention	Same endpoints, different interior z ; shared training or evaluation settings	Independent contribution of z ; Apps. B.1, F.2
Cosh learning	Scratch or shared-start continuation	Analytic allocation versus Geo under a matched recipe; grid/endpoint treatment stated per experiment	Learned extrapolation; Apps. E.1, D.5
TailSpline deployment	Frozen; zero training	Static closed-form table versus MrPro; shared band, endpoints, gain and inputs	Two-model method results; App. H
BM natural QA	Frozen; zero training	Symmetric boundary construction versus matched MrPro	Natural-response F1; App. G.5
Matched adaptation	Explicit adapter continuation	Paired table/adapter recipes with declared long-phase exposure	Learned complete-answer readout; App. F.1

A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (10)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. Thus $c_{\omega\nu}$ is an intrinsic measure of overlap between the two positional subspaces.

Theorem 4 (Spectral budget identity). *For the block-whitened Gram matrix Γ with diagonal blocks I_2 , $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$. Hence its Rényi-2 effective rank is given by (5).*

Proof of Theorem 4. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 effective rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 1. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (11)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (12)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 . Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (13)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (14)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (15)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1} M) + O(\epsilon^6) = \frac{19}{12600} (x^2 - y^2)^2 + O(\epsilon^6), \quad (16)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega \Delta)}}{\omega} \longrightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{2\overline{\cos(\omega \Delta)}}{\omega^2} \longrightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (17)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. The separation weights p therefore determine which polynomial directions survive centering. $\square$

Directional overlap and feature scale. For the original, unwhitened pair on $[0, L]$, put $x = \omega L$ and $\text{sinc } x = \sin x/x$. The self-Gram eigenvalues are

$$\lambda_{\pm}(S_{\omega}) = \frac{1 \pm |\text{sinc } x|}{2}, \quad \lambda_{-}(S_{\omega}) \sim \frac{x^2}{12}, \quad \kappa(S_{\omega}) \sim \frac{12}{x^2} \quad (x \rightarrow 0).$$

Thus the slow pair has a poorly scaled second direction before normalization. Block whitening removes this within-pair scaling and measures overlap between positional directions. Raw feature energy, coefficient magnitude, and numerical precision determine how strongly those directions can contribute to a model.

The standard $u_k = k/K$ grid gives 23 slow pairs in Figure 5; the endpoint-inclusive grid used in Figure 6 gives 24 at $K = 64$. Both figures use the uniform separation measure.

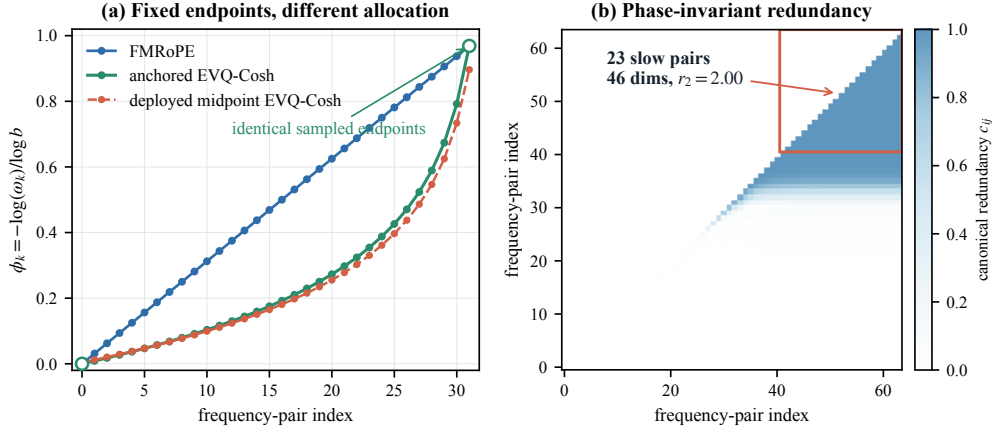


Figure 5: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

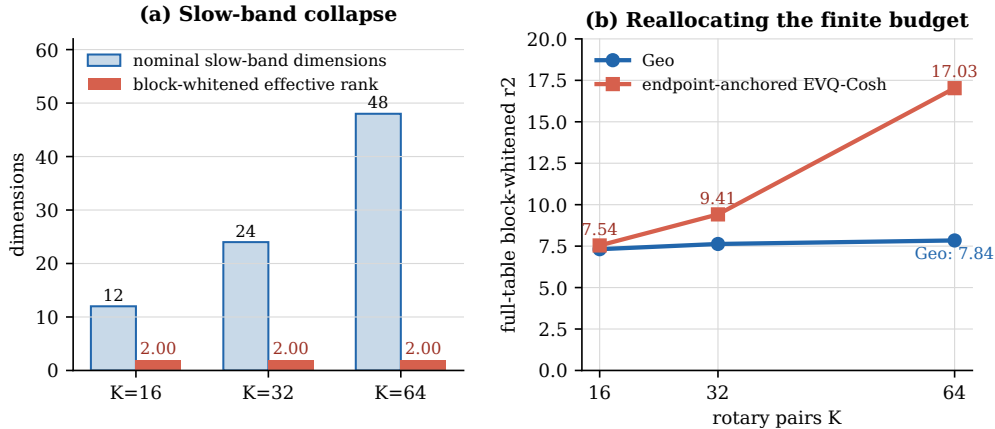


Figure 6: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows.

A.2 FINITE-WINDOW OVERLAP IN THE IDENTIFICATION EXPERIMENT

The slow-limit proposition has a finite-window counterpart that can be calculated directly on the identification grid. Set $b = 256$, $K = 32$, $L = 256$, and retain the n slowest frequencies of $\omega_j = b^{-j/K}$. Their dimensionless frequencies are $\omega_j L = 1.1892, 1.4142, 1.6818, 2, \dots$ in slow-to-fast order. The smallest is already greater than one, so we evaluate the full cross-Gram in (10) without a small-phase approximation.

Table 5: **Finite-window positional concentration on the paired-training grid.** The separation measure is uniform on $[0, L]$. The nominal coordinate count is $2n$; r_2 is the effective rank of the block-whitened Gram.

Slowest pairs n	Largest ωL	Coordinates	r_2
2	1.41	4	2.00030
4	2.00	8	2.00361
6	2.83	12	2.02263
8	4.00	16	2.11474
12	8.00	24	2.89256
16	16.00	32	4.43339

For $n = 8$, the largest two Gram eigenvalues are 7.9681 and 7.5752, accounting for 97.1% of the trace; the third is 0.4246. As faster pairs enter, additional positional directions acquire mass. This is the finite-window overlap represented in Fig. 7. The accompanying `figs/make_story_figures.py` checks the $n = 8$ closed-form calculation against independent 256-node Gauss–Legendre quadrature and also recovers $r_2 = 2.00013$ for the 23 slow pairs of the $b = 500,000$, $K = 64$, $L = 4096$ grid. The machine-readable calculation is in `figs/finite_window_geometry.json`.

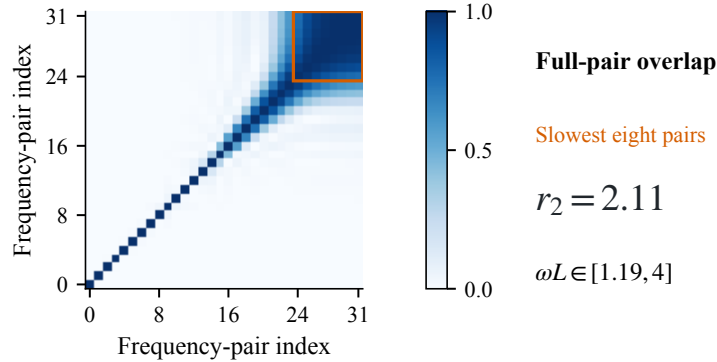


Figure 7: **Slow pairs share positional directions.** Full-pair canonical overlap on $b = 256$, $K = 32$, $L = 256$ under uniform separations. The slowest eight pairs have $r_2 = 2.11$ despite $\omega L \in [1.19, 4]$.

A.3 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

Proposition 5 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (18)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (19)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (20)$$

so every entry of the cross-Gram in (10) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_\star((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (19) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k\Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 5 couples maximal rank on $[0, L]$ with recurrence beyond that interval. Its orthogonality also depends on the separation measure: weighting the same lattice by the triangular causal histogram introduces off-diagonal Gram terms. Thus interval coverage and pairwise rank capture different features of a positional basis.

A.4 TWO STATIC ORDERING COUNTEREXAMPLES

Two explicit examples show how phase coupling and interval length change the ordering of frequency tables.

Cosine-only collision can choose the lower-rank table. Take $K = 4$, $L = 8\pi$, and the two descending frequency tables

$$\Omega_A = (1, 5/8, 3/8, 1/4), \quad \Omega_B = (1, 6.01/8, 4.01/8, 1/4).$$

They share both endpoints. Define $C_{\cos}$ as the mean squared normalized cosine–cosine overlap over distinct pairs. For Ω_A , every cosine is an integer harmonic on $[0, L]$, so $C_{\cos}(A) = 0$ exactly. Mixed harmonic parities nevertheless produce nonzero sine–cosine cross terms. For Ω_B , the small perturbation of an even-harmonic table gives

$$C_{\cos}(A) = 0 < 1.38725 \times 10^{-5} = C_{\cos}(B), \quad (21)$$

$$r_2(A) = 6.29478 < 7.99971 = r_2(B). \quad (22)$$

Thus a strict cosine-only preference can have the opposite full-subspace rank ordering. Both values follow directly from Eq. (10).

Full-subspace collision can reverse across lengths. Now use $K = 4$, $L = 2\pi$, and the fixed tables

$$\Omega_A = (1, 0.99, 0.02, 0.01), \quad \Omega_B = (1, 2/3, 1/3, 0.01).$$

Again the endpoints agree. Writing C_M for mean phase-invariant collision under the uniform measure on $[0, M]$, Eq. (10) gives

$$C_L(A) = 0.538996 < 0.631721 = C_L(B), \quad (23)$$

$$C_{2L}(A) = 0.384445 > 0.204995 = C_{2L}(B), \quad (24)$$

$$C_{4L}(A) = 0.345188 > 0.033506 = C_{4L}(B). \quad (25)$$

The close pairs in A retain high overlap as the interval grows, while the more evenly spread frequencies in B become easier to distinguish. The example makes the interval dependence of the geometry explicit.

All frequency entries and measures are specified above. The accompanying `figs/verify_explicit_geometry.py` evaluates the closed-form Gram and checks it against independent 128-point Gauss–Legendre quadrature; the largest discrepancy in the displayed metrics is below 4×10^{-14} . These finite basis examples complement the trained allocation comparisons and the weight–table crossings.

Table 6: Weights co-adapt to their training-time frequency table. Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head–query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic ($K = 32$, $b = 500,000$, $L = 512$). Static rank uses equal weights on integer separations $0, \dots, 511$, distinct from the continuous measure of the main geometric proposition. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

A.5 POST-HOC TRANSPLANT OBSTRUCTION

Theorem 6 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta) B = \mathcal{R}_\Omega(\Delta)$ on an open interval containing zero, then Ω' and Ω have the same frequency multiset up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Proof of Theorem 6. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace, including across its original two-dimensional blocks. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

A.6 INTEGER-POSITION KERNEL EQUIVALENCE AND ITS BOUNDARIES

Corollary 2 applies to the full bilinear kernel on the $2K$ -dimensional rotary space. Equality for all q, k is matrix equality. At $d = 0$, $A^\top B = I$; at $d = 1$ the two rotation matrices are similar. Their complex eigenvalues are $e^{i\omega_k}, e^{-i\omega_k}$ including multiplicities. On $(0, \pi)$ the positive-angle eigenvalue identifies each frequency uniquely, proving necessity even with repeated frequencies. For sufficiency choose a block permutation P with $\mathcal{R}_\Lambda(d) = P \mathcal{R}_\Omega(d) P^\top$, and set $A = B = P^\top$. This holds for every integer d , including negative distances. This is the discrete specialization of Theorem 6, using standard similarity rather than a new classification of position encodings.

Counterexamples and excluded extensions. A single trace is insufficient: $(\arccos .9, \arccos .1)$ and $(\arccos .7, \arccos .3)$ have the same one-step trace, but different rotation spectra and different two-step traces. Outside $(0, \pi)$, replacing ω with $\omega + 2\pi$ preserves every integer-position rotation while changing rotations at fractional positions. Frequencies at zero or π introduce real eigenvalue blocks; a partial rotary operator adds an identity subspace. The stated corollary excludes these cases. Nonlinear Q/K normalization need not commute with a candidate linear compensation. Neither exact non-equivalence nor a slot-permutation loss proves that retraining cannot adapt. The supplied CPU checker verifies compensation, repeated frequencies, same-trace/different-spectrum examples, integer aliasing, and the limits of a scalar gain; it measures no model performance.

Approximate compatibility within a geometric family. The exact kernel result leaves room for approximate recovery of a model’s behavior. In the historical 50M crossing, three geometric controls were derived from the Cosh table without fitting model loss: matching its slow endpoint ($b = 331.6K$), matching its log span (377.7K), and minimizing log-frequency RMS within the geometric family

(8.06K). On Cosh-trained weights these give PPL 21.69, 22.14, and 9.63, compared with 23.05 for the original Geo table and 7.16 for the training Cosh table. A coarse geometric approximation can therefore recover substantial frozen compatibility. On Geo-trained weights the respective PPLs are 7.30, 7.16, and 45.63 (original Geo: 7.14). These historical tables also change support, so the fixed-support training and mature-model controls supply the separate interior-allocation estimates.

A.7 SLOT ASSIGNMENT AND CONDITIONAL PREDICTION ERROR

Let $\mathcal{R}(\Omega; M, D)$ denote the risk of a fixed model M on a fixed evaluation set D when frequencies Ω are assigned to labeled rotary slots. This domain includes unsorted assignments. It differs from the strictly ordered allocation domain in §2, where the full multiset recovers the sorted table.

Proposition 7 (Prediction error on an observed permutation pair). *Suppose a pair $\Omega, \pi\Omega$ has risk gap $\Delta_R = |\mathcal{R}(\Omega; M, D) - \mathcal{R}(\pi\Omega; M, D)| > 0$. For every permutation-invariant feature map Φ and every scalar predictor h ,*

$$\max_{\Xi \in \{\Omega, \pi\Omega\}} |h(\Phi(\Xi)) - \mathcal{R}(\Xi; M, D)| \geq \Delta_R/2.$$

Proof. Invariance gives the same prediction $c = h(\Phi(\Omega))$ for both assignments. The triangle inequality bounds Δ_R by the sum of the two prediction errors, and hence by twice their maximum. $\square$

The risk-gap premise is empirical. It does not follow from unequal projection norms or from Theorem 6. Synchronously transforming the Q/K blocks compensates the permutation exactly by (68); changing the table while freezing those projections measures a different intervention. In the measured pair, OLMo-1B PG-19 tail NLL changes from 3.10423 to 6.86493, and Qwen-2.5-1.5B 64K core-four macro changes from 0.7000 to 0.0000.

Discrete orbit-count sensitivity. In the archived scale-orbit construction at factor four, one-ULP float32 jitter changes the exact orbit count from 6 to 64, while the 16K tail-NLL difference (jitter minus exact) is -0.000013 (paired interval $[-0.003682, 0.003715]$) and the 16K core-four score remains zero. This rejects the interpretation that a large increase in this exact algebraic count necessarily produces an appreciable loss change on the tested finite horizon. It does not rule out a predictor that maps different counts to the same risk. Discrete-period design, as studied by Resonance RoPE (Wang et al., 2024a), should be distinguished from this exact-ratio sensitivity test.

A.8 FINITE-PHASE OPERATOR CHANGES

For $\delta = (\omega' - \omega)\Delta$, the second-order cosine remainder is bounded by $|\delta|^3/6$. The series converges for all δ , but a low-order approximation need not be accurate at large phase increments. An exact finite-phase operator bound is

$$\|\mathcal{R}_{\Omega'}(\Delta) - \mathcal{R}_{\Omega}(\Delta)\|_2 = 2 \max_j \left| \sin \frac{(\omega'_j - \omega_j)\Delta}{2} \right| \leq \max_j \min\{2, |\omega'_j - \omega_j| |\Delta|\}. \quad (26)$$

A.9 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

Let $S_\rho(t) = \int_t^1 \rho(\phi) d\phi$. The Cosh density-design objective is

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \int_0^1 S_\rho(t)^2 dt, \quad \int_0^1 \rho = 1, \quad \rho \geq 0. \quad (27)$$

With unit mass, the first term favors spreading frequencies. Moving density toward slower exponents increases tail mass over the thresholds it crosses; the second term penalizes that concentration. These terms specify the extrapolation transport prior.

Theorem 8 (Unique density minimizer). *For $\alpha > 0$ and $\beta \geq 0$, the unit-mass nonnegative L^2 minimizer of Eq. (27) is uniquely $\rho_\tau(\phi) = \tau \cosh[\tau(1 - \phi)] / \sinh \tau$, with the uniform limit at $\tau = 0$.*

This subsection proves the theorem directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (28)$$

the allocation functional is (27), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha\rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (29)$$

Existence. View $\mathcal{C}_{\text{app}}$ as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e.}, \int_0^1 \rho = 1\}$. The $\alpha\|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_\phi^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (30)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (31)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (32)$$

since differentiating $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha\rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha\rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (31) and (32) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (33)$$

which establishes Theorem 8 and the density used in §5.2. Mass normalization enters the boundary condition through $g'(0) = \int_0^1 \rho = 1$. Positivity holds on the closed interval since $\rho_\tau(\phi) \geq \rho_\tau(1) = \tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_\tau(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (33).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the objective is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (34)$$

The kernel is the inverse of $-d^2/d\phi^2$ with Dirichlet condition $g(0) = 0$ and Neumann condition $g'(1) = 0$. Thus the tail term is an H^{-1} -type energy with these mixed boundary conditions: $\langle \rho, g \rangle = \int_0^1 |g'|^2 = \int_0^1 S_\rho^2$. This classifies the stated design prior without changing its empirical operating rule.

The strictly convex functional on $L^2([0, 1])$ therefore has the displayed stationary density as its unique constrained minimizer.

The Cosh solution has an analytically invertible CDF, so its quantiles turn the continuous allocation directly into a finite frequency table.

A.10 THE DENSITY OBJECTIVE IN INTERVAL COORDINATES

For a strictly positive density, let $Q = F^{-1}$ and $h(u) = Q'(u) > 0$ with $\int_0^1 h = 1$. Then $\rho(Q(u)) = 1/h(u)$ and $S_\rho(Q(u)) = 1 - u$. Changing variables in the declared density objective gives the exact identity

$$\mathcal{C}_{\text{app}} = \frac{1}{2} \int_0^1 \left[\frac{\alpha}{h(u)} + \beta(1-u)^2 h(u) \right] du.$$

Thus Cosh can equivalently be viewed as an interval-allocation design. Its quantile derivative is

$$h_\tau(u) = \frac{\sinh \tau}{\tau \sqrt{1 + (1-u)^2 \sinh^2 \tau}}.$$

This equivalent interval formulation connects density allocation to frequency spacing. TailSpline instead designs increments of native-relative displacement; the two objectives address learning and frozen deployment.

A.11 SINGLE-CROSSING BUDGET SHIFT

Lemma 9 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (35)$$

Proof of Lemma 9. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (35). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. $\square$

A.12 BALANCE OF THE DENSITY OBJECTIVE

The density objective $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 10 (Density-objective balance). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (36)$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the Cosh ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the Cosh density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (37)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (38)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (36). $\square$

An immediate corollary follows.

Corollary 11 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (39)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity provides a closed-form evaluation of the allocation energy.

A.13 ALLOCATION STRENGTH AND A REFERENCE RULE

The Cosh family varies with the strength τ . A reference used in the full-RoPE text experiments relates this strength to head width and training length:

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \frac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (40)$$

The reference sets $c = 1$ before training. Table 12 records the strength, grid, and setting procedure used in each experiment.

Local scaling calculation. The scaling argument uses four explicit modelling assumptions: (i) the Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (41)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (42)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1 - \phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (43)$$

The stiffness itself also has an exact expression:

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \arctan(\sinh \tau)}{\tau^2} - 1 \right]. \quad (44)$$

Indeed, unit mass reduces its defining integral to $\int \rho_\tau^{-1} - 1$, and integrating sech gives Eq. (44). The quartic term is a local expansion: at $\tau = \sqrt{2}, 2, 4$, it exceeds the exact stiffness by factors approximately 1.52, 1.97, 3.52. It is not a finite- τ accuracy guarantee for the operating rule.

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives the leading small- τ balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (45)$$

This calculation yields the scaling factor $d_{\text{head}}/\sqrt{L}$ with a local coefficient determined by $Q_1(L, b)$ and λ . The experiments evaluate the exact Cosh quantiles at the protocol’s chosen strength.

Reference and neighboring strengths. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at $0.75\times$, $1.25\times$, or $1.5\times$; the independent exact-range factorial

tests $0.75\times$, $1.00\times$, and $1.25\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entriypoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. The reference and its neighboring strengths give a direct comparison of allocation intensity across the tested configurations.

A.14 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. We compare the measures in Wasserstein distance and approximate the density with the associated quantile-cell histogram ρ_K .

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (46)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (47)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (48)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (49)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoﬀ, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (50)$$

This expansion quantifies how finite exponent samples approximate a continuous allocation under a smooth distortion. For endpoint-anchored comparisons, $K = 2$ leaves no interior entries and the geometric and Cosh grids coincide.

Large- τ regime. At large τ , $m_\tau = \tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B CONTROLLED ALLOCATION EXPERIMENTS

B.1 EXACT-RANGE CONTROL AT 151.9M

Paired controls. The two trained tables have identical frequency endpoints. Within each seed pair the arms share: model architecture (151,898,880 parameters); trainable initialisation; training length

256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Matched geometric control (Geo):* fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Writing Φ_τ for the inverse CDF in (9), let $q_k=\Phi_\tau((k + \frac{1}{2})/K)$, $s_k=(q_k - q_0)/(q_{K-1} - q_0)$, and $R=((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{Geo}} = \exp\left(-R\frac{k}{K-1}\right). \quad (51)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$. The anchors are common end positions selected with Python’s `random.Random(20260723)`: the eligible validation stream is divided into 32 equal-width bins, with one end position drawn per bin and a 2,048-token guard before the next bin. At anchor e , length L uses validation tokens $[e-L, e)$ and scores the final 128 targets $[e-128, e)$. Longer contexts therefore add earlier text around the same target tokens; the longest windows at distinct anchors are disjoint.

Exact range transformation.

$$R(L) = \frac{K-1}{K} \log L, \quad \omega_k^{\text{target}}(L) = \exp[-R(L)z_k].$$

Here $z_k = k/(K-1)$ for Geo and the anchored Cosh shape for its paired arm. The fixed condition uses $L = 256$ at every evaluation length; the target-matched condition substitutes the declared evaluation length. Both conditions use amplitude one.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Across all three seeds, fixed training range favors Cosh at every extended length, whereas target-matched range favors the geometric control. The paired design thus shows both an interior-allocation effect and a change in preferred allocation when the same trained weights are evaluated under a retargeted range.

Table 7: Exact-range anchored EVQ-COSH minus the matched geometric control tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

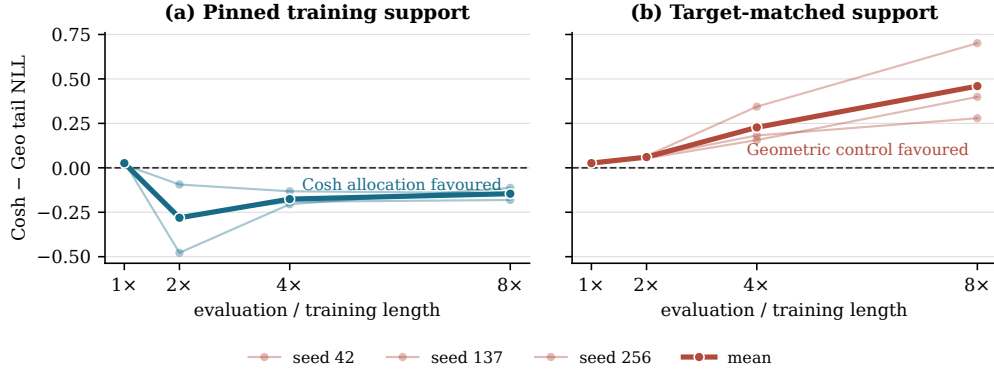


Figure 8: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

B.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired Geo and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

For each training table Ω , the derived table uses the following label-free residual construction. Set $L_0 = 256$, $\Delta_j = 2j$ for $j = 0, \dots, 127$, and $p_j = (2L_0 - 4j - 1)/(L_0(L_0 + 1)/2)$, the grouped causal-separation weights. Let X_k have columns $\sqrt{p_j} \cos(\omega_k \Delta_j)$ and $\sqrt{p_j} \sin(\omega_k \Delta_j)$, and let X_{-k} collect all other pairs. Define

$$U_k = \frac{\|X_k - X_{-k} X_{-k}^\dagger X_k\|_F^2}{\|X_k\|_F^2}, \quad \bar{U}_k = \frac{U_k - \min_i U_i}{\max_i U_i - \min_i U_i}, \quad w_k = (1 - \bar{U}_k)^2.$$

The pseudoinverse uses relative singular-value cutoff 10^{-10} ; the zero-span convention is $\bar{U}_k = 0$. The frequency blend is $\omega'_k = \omega_k(1 - 3w_k/4)$, cast to FP32 with its endpoints set exactly to $\omega'_0 = \omega_0$ and $\omega'_{K-1} = \omega_{K-1}/4$. Both derived tables use $g = 1 + 0.1 \log 4$ and remain unchanged across the 512- and 1,024-token evaluations.

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. Both seeds exhibit the same preference for the runtime table derived from their training allocation.

Table 8: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	Geo-derived table	Anchored-EVQ-COSH-derived table
Geo-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

B.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at $0.75\times$, $1.00\times$ (the formula point), $1.25\times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Evaluation data and windows. Training and evaluation use the separate train and validation splits of cached wikitext-2-raw-v1, tokenized by GPT-NeoX-20B. Empty text rows are removed, remaining rows are stripped and concatenated without extra special tokens, and each split is repeated/truncated independently: 8,388,608 training tokens and 1,048,576 evaluation tokens. The data revision of that legacy Arrow cache is not recorded. For each arm, NumPy RandomState(9999) samples four distinct starts at each of $L, 2L, 4L, 8L$ in that order. Full next-token cross-entropy scores $4(rL - 1)$ targets at length rL , averaging the four chunk losses before exponentiation. All arms share these offsets. The split is distinct from training, but repeated validation text and potentially overlapping windows are not independent documents; uncertainty is computed over the structural configurations as specified below.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75\times$ 6.202858 (494.16); EVQ-COSH $1.00\times$ 6.202094 (493.78); EVQ-COSH $1.25\times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5\times$ arm is better; at formula $\tau=8$ the $0.75\times$ arm is. The improving direction points toward the intermediate strengths at both tested extremes.

The factorial varies architecture, base, and training length on a repeated WikiText-2 stream (Merity et al., 2017). Its configuration-level contrasts complement the longer FineWeb-Edu training comparison in §B.1.

The joint view shows simultaneous training-window and OOD improvements at multiple operating points. The measured trade-offs vary across allocation shapes, strengths and model configurations.

Table 9: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. B).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH 0.75 $\times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH 1.25 $\times$ – Geo	−0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

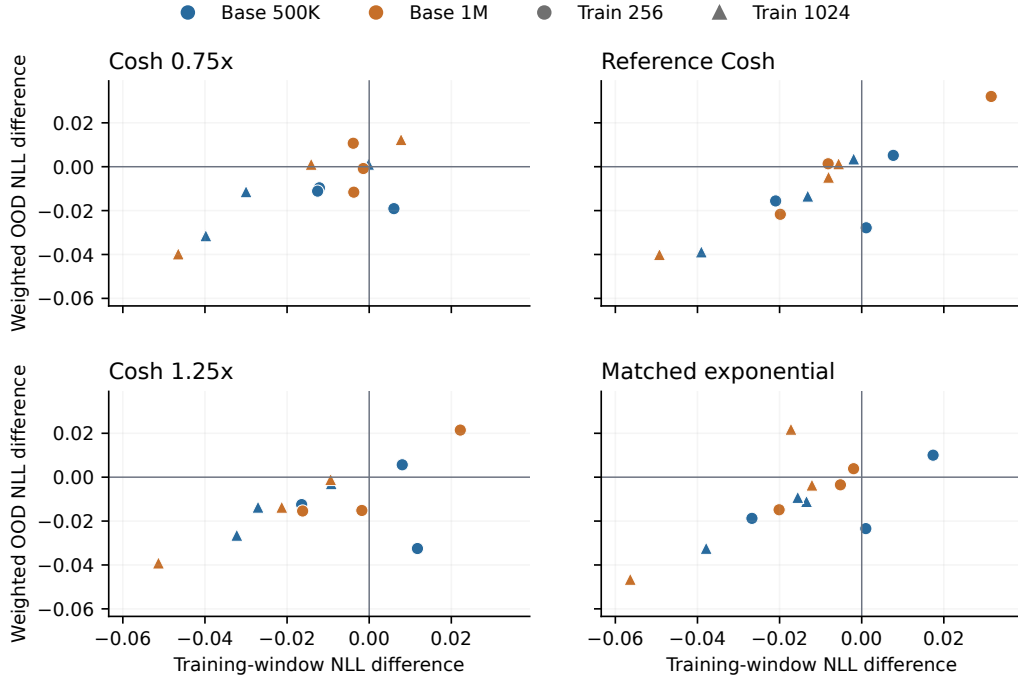


Figure 9: **Allocation changes training-window and extended-length loss jointly.** Each point is one of the twelve prespecified M4 configurations, averaging three paired seeds; all four nonuniform main arms are shown. Both axes are differences from the matching geometric arm, so negative values improve loss. OOD weights are proportional to $\log_2(r + 1)$ for $r \in \{2, 4, 8\}$, as in the original protocol. The horizontal coordinate is reconstructed from recorded four-decimal PPL; the vertical coordinate uses the recorded full-precision weighted NLL. The configurations form a prespecified design grid.

B.4 FINITE DEPLOYMENT INTERVENTIONS

The permutation and finite- C_2 results in Appendix A.7 concern fixed-weight interventions. The C_2 approximation has its largest coordinate residual at slot 19. With zero-based indexing and native $K = 64$ geometric frequencies, that slot makes 13.25 turns in OLMo’s 4096-token window ($b = 500,000$) and 86.30 turns in Qwen’s 32768-token window ($b = 1,000,000$). Neither is the native one-turn boundary. A transition in an installed profile and the one-turn boundary of a native table are distinct quantities.

Three-band profiles retain a fast band, fully scale a slow band, and distribute the shift across an intermediate band. The cutoffs specify a construction; performance is measured with the checkpoint

and gain held explicit. On the OLMo six-task confirmation panel, BM improves 16K macro from 2.78% to 51.32% relative to MrRoPE-Pro. On the separate 631-prompt natural-QA long-input stratum it improves task-equal F1 from 21.62% to 25.44%. Detailed model contrasts and scoring protocols are in Appendix G; the exploratory twelve-profile panel is documented in Appendix J.6.

C LEARNED-FREQUENCY COMPARATOR

Table 10: Learned-frequency comparator (approximately 151.9M, $L_{\text{train}}=128$, FineWeb-Edu, 128 $\rightarrow$ 8K). Geo denotes the midpoint geometric grid. Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). The learned-inverse-frequency arm has one parameter per pair, shared across layers and heads.

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm learns one inverse frequency per rotary pair, shared across layers and heads. This is the learned-table parameterization also studied by Karypis et al. (2026). Its positional-parameter learning-rate sweep uses 10 $\times$ and 100 $\times$ the base rate; the stronger 100 $\times$ setting gives 455.3 PPL. The learned table and the fixed Cosh allocation both improve on Geo in this protocol.

These inverse frequencies are learned without fixing the sampled endpoints, so this comparison varies realized support as well as interior placement. This parameterization comparison uses 15M training tokens.

C.1 HISTORICAL NATIVE-GRID SHAPE CONTROL

A separate three-seed study compares the native standard-grid RoPE reference with a span-matched native-grid Cosh table. It uses 151,898,880 parameters, base 500,000, head dimension 64, training length 128 and 14,974,976 tokens. The same study compares additional analytic and kernel-derived shapes at matched span and RMS deformation. Selection uses 16 anchors and evaluation 32 disjoint test anchors. This is a short-training shape study, separate from the 500M-token fixed-support replication. The recorded seed-level 95% intervals for Cosh-minus-Std-RoPE NLL at 512/1K/2K/4K/8K are

Table 11: Historical native-grid three-seed mean tail NLL. The complete length response retains the training window.

Table	128	256	512	1K	2K	4K	8K
Native Std-RoPE	5.5046	5.5026	5.6563	5.8456	6.0530	6.2566	6.3928
Native Cosh	5.4645	5.4473	5.5434	5.6966	5.8457	6.0664	6.2935

[−0.200, −0.026], [−0.183, −0.115], [−0.241, −0.174], [−0.260, −0.121], and [−0.141, −0.058]. All three seeds favor Cosh at each of those lengths. This supports the conclusion that the historical extrapolation improvement is not confined to a midpoint geometric comparator. The same report also includes other useful nonuniform shapes. The source archive preserves the reported values and historical source identity in `figs/allocation_value_inputs.json`; the original training streams are separate.

D EXPERIMENTAL DETAILS

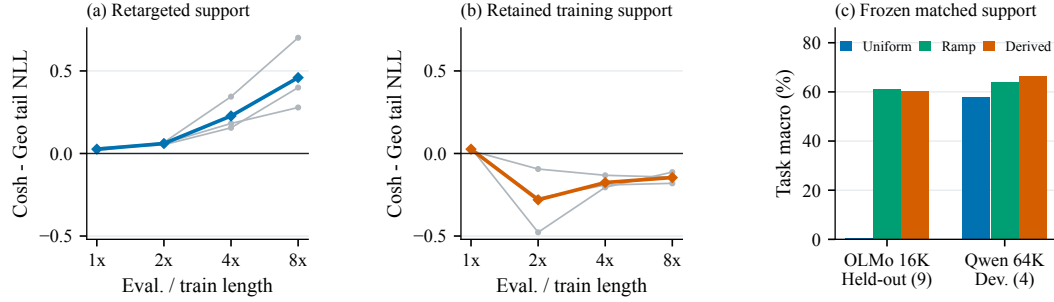


Figure 10: **Allocation gains depend on the support policy.** (a,b) Cosh-minus-Geo tail NLL for the same three paired 151.9M models: retargeted support ($b = L_{\text{eval}}$) versus retained training support ($b = 256$). Thin lines show seeds; thick lines their means; $1\times = 256$ tokens. Geo is the matched geometric control. (c) Frozen matched-support tasks: OLMo’s nine held-out and Qwen’s four development tasks, 20 inputs/task, with shared endpoints, amplitude, weights and inputs within each model.

D.1 EXPERIMENT ORGANIZATION

This section gives the training, continuation, and video protocols. Apps. B and F detail the fixed-range and mature-model experiments.

D.2 TABLE AND OPERATOR IDENTITIES

RoPE, Geo and Native. Geo is a geometric RoPE table with $\omega_k = b^{-u_k}$. For standard $u_k = k/K$ and midpoint $u_k = (k + 1/2)/K$ grids at the same base,

$$\omega_k^{\text{mid}} = b^{-1/(2K)} \omega_k^{\text{standard}}.$$

Both are geometric RoPE with identical normalized allocation z and log-span R ; the midpoint offset changes a in Eq. (1). For $b > 1$, the factor is below one, giving uniformly slower rotation at the same token separation. The midpoint Cosh family’s $\tau \rightarrow 0$ limit is this midpoint Geo. Native identifies the inherited table of a specified released model. Each experiment retains its actual grid and baseline; geometric equivalence of the family does not by itself quantify a task-score difference.

Cosh shape and finite installation. For $\tau > 0$, the Cosh density is decreasing, so Q_τ is convex. Affine endpoint anchoring on equally spaced quantile nodes gives

$$z_{\tau,k} = \frac{Q_\tau(u_k) - Q_\tau(u_0)}{Q_\tau(u_{K-1}) - Q_\tau(u_0)} \leq \frac{k}{K-1}, \quad \omega_{\tau,k} \geq \omega_{\text{Geo},k}. \quad (52)$$

The inequality follows from convexity relative to the endpoint chord.

The family in Eq. (9) uses $\phi_\tau(u) = 1 - \text{arcsinh}((1-u) \sinh \tau)/\tau$. Grid choice and optional affine anchoring determine the installed tensor. At fixed endpoints, the anchored shape satisfies Eq. (52); a midpoint deployment without anchoring also changes the sampled range. Results are assigned to these actual installations, not to the family name alone.

Extension operators. Official-equation YaRN, the MLA wavelength blend in Appendix E.1, and the repository fixed-index ramp in Eq. (53) are distinct transformations. The last two appeared under historical YaRN labels in evaluation outputs; this manuscript names their actual operators. Paired gains with a shared operator remain evidence for that composition, and are not relabeled as official YaRN comparisons.

D.3 MODELS AND TRAINING SETTINGS

Table 12 records allocation and deployment parameters; Table 13 gives training settings. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

The front-of-appendix protocol glossary (Table 4) separates identification, learning, adaptation and frozen deployment. Additional shape, placement, architecture and video studies retain their local sample units and endpoints in their respective sections.

Table 13 lists the training hyperparameters shared across the supporting text experiments.

Table 12: **Allocation protocols, parameter bounds, and float32 tensor SHA-256 digests. Top:** Cosh-family settings across reported experiments. Midpoint means $u_k = (k + 1/2)/K$; standard means $u_k = k/K$; anchoring fixes sampled endpoints after quantile evaluation. **Bottom:** Parameter bounds and float32 tensor hashes for static deployment profiles. MrRoPE-Pro parameters follow discrete turn cutoffs $\beta = 32, \alpha = 1$ with fast cutoff $l = \max\{j \mid \text{turns}_j > 32\}$, slow cutoff $h = \min\{j \mid \text{turns}_j < 1\}$, and transition width $n = h - l$. FullLagP2 evaluates all 32,768 integer lags, rather than the earlier subsampled grid.

Part A: Cosh-Family Pre-training and Adaptation Settings

Protocol	τ	Grid	Setting convention
151.9M fixed-range	4	Anchored midpoint	Fixed $64/\sqrt{256}$ reference
50.9M factorial	Varies	Anchored midpoint	Three preassigned multiples of $d_{\text{head}}/\sqrt{L_{\text{train}}}$
50M crossing	2.83	Midpoint	Reported training-table setting
432M MLA	1.414	Midpoint	Pre-specified architecture-specific setting
454M composition	1.5	Midpoint	Reported training recipe
750M continuation	1.5	Anchored inclusive	Fixed continuation recipe; $u_k = k/(K - 1)$
Llama-3-8B	1.414	Midpoint	Fixed adaptation recipe; approximately $128/\sqrt{8192}$
OLMo-2	2	Standard	Fixed $128/\sqrt{4096}$ reference
129.6M Video DiT	1.5	Midpoint	Best tested value in the video sweep

Part B: Static Deployment Profiles, Bounds, and Tensor SHA-256 Digests

Profile / Architecture	Scale s	Bounds / Lags	Float32 Tensor SHA-256 Digest
MrRoPE-Pro (Qwen2.5)	4	$l=23, h=40, n=17$	33cbe3a4 0994ac2a 79126a14 ce302828 67bd6d49 b74c1ada 4d4cce8a 7e76016f
MrRoPE-Pro (OLMo-2)	4	$l=14, h=32, n=18$	ec9c5497 77f7a94c 73a1cc2d b5f22fef be202a7e 083e454f 2510034f 2cec5dfb
FullLagP2 (Qwen2.5)	4	32,768 integer lags	ecd0c280 a11788e0 c4a869a3 d1628809 64ba3f37 1f589f7e 2f5ae471 d461096b
Boundary-Matched (OLMo-2)	4	$s=4, g=1.1386$	c80626f8 f9280854 577fd1e7 94439300 09a69370 65366adb 8eee5f62 ce7807a6
Native RoPE (OLMo-2)	1	$L_0=4096, b=500K$	9229ec6a 6a7ce5c1 750273bc dbce9789 5a74a920 2722f14b 6ca2328e ce5ee98e
TailSpline (Llama-3)	4	$l = 18, h = 35, n = 17$	See paired source identities, App. H
TailSpline (OLMo-2)	4	$l = 14, h = 32, n = 18$	See paired source identities, App. H

Within the scratch and continuation comparisons specified in this section, the architecture, optimizer, data order, and training loop are fixed, and the intervention is the initialized inverse-frequency table. The released-checkpoint 1.485B comparison in Appendix F.1 uses separate trainer implementations, as detailed there. The scratch-text experiments use base $b=500K$ unless stated otherwise.

Table 13: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 14 (750M: micro 7, accumulation 2), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

D.4 COMPOSITION WITH A FIXED-INDEX RANGE OPERATOR

The 454.2M decoder has 24 layers, width 1024, 16 heads, $d_{\text{head}} = 64$, and base $b = 500,000$. Geo ($\tau = 0$) and EVQ-COSH ($\tau = 1.5$) use midpoint frequency grids and train from initialization for 100M tokens per arm at length 2048, using FineWeb-Edu with a 10% synthetic-passkey mixture and seeds 42, 123, 7. The same inference-time operator is then applied to both trained tables.

For $k = 0, \dots, K - 1$, define

$$\ell = \lfloor 0.20K \rfloor, \quad h = \lfloor 0.90K \rfloor, \quad t_k = \text{clip}\left(\frac{k - \ell}{h - \ell}, 0, 1\right), \quad r_k = t_k^2(3 - 2t_k).$$

The repository’s fixed-index, YaRN-style smooth-ramp operator installs

$$\mathcal{R}_s(\omega_k) = \frac{\omega_k}{s^{r_k} T(s)^{r_k/2}}, \quad T(s) = 1 + 0.07 \log_2 s. \quad (53)$$

Here $K = 32$ and $s = 8$, giving $(\ell, h, T) = (6, 28, 1.21)$. The temperature factor is folded into the frequency divisor; cosine/sine amplitude stays at $g = 1$. Equation (53) specifies the operator used in this experiment, separately from the index-linear YaRN comparisons in Appendix G.8. Table 14 reports the three-seed comparison.

Table 14: Matched fixed-ramp composition on the 454M three-seed experiment. $\mathcal{R}_8$ is defined in Eq. (53). PPL uses full-sequence scoring; PK is teacher-forced NLL-gap retrieval. Entries are three-seed means; PK@8K also gives the standard deviation.

Table	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41±5%	57%	51%	161.9	253.2
Geo+ $\mathcal{R}_8$	61±3%	59%	51%	82.9	157.7
EVQ-COSH	53±8%	63%	50%	150.3	229.5
EVQ-COSH+ $\mathcal{R}_8$	100±0%	79%	68%	70.9	107.5

D.5 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, fixed-length autoregressive answer-token exact match is 0% versus 77.5%. The EVQ-COSH continuation uses $\tau = 1.5$, $K = 32$, and the endpoint-inclusive

quantiles $u_k = k/(K - 1)$. Writing q_τ for the inverse CDF in Eq. (9), the installed frequencies are

$$\omega_k = \exp \left[-\frac{K-1}{K} \log b q_{1.5} \left(\frac{k}{K-1} \right) \right], \quad b = 500,000.$$

This retains the Geo endpoints. Both arms are full-parameter continuations from the shared checkpoint. The autoregressive probe generates exactly the tokenized passkey plus its closing marker and compares every generated token ID with that expected sequence. It does not test a subsequent terminal EOS; the EOS-complete endpoint is the separate OLMo study in Table 25.

Table 15: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials per length. The pooled AR row combines 2K, 4K and 8K (120 trials: Geo 80/120, EVQ-COSH 111/120).

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Answer-token exact @8K	0%	77.5%
Pooled answer-token exact (2K/4K/8K)	66.67%	92.5%

D.6 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau = K_t/\sqrt{T} = 2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ (0.53 $\times$), which selects the supporting video configuration.

D.7 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames (4 $\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 16 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames under bidirectional 3D RoPE.

Table 16: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

D.8 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 6 separately measures incompatibility after a frozen post-hoc swap.

D.9 ALLOCATION UNDER GQA AND MLA COMPRESSION

A seed-42 125M study compares five attention configurations, each trained for 100M FineWeb-Edu tokens at length 2048 with 2% passkey mix. Within each configuration Geo and Cosh ($\tau = 1.414$) use the matched training recipe. Table 17 preserves the complete configuration comparison from the recorded experiment report.

Table 17: **Allocation effects across attention configurations.** Paired Geo/Cosh PPL at each length; the last column is the reported teacher-forced passkey retrieval aggregate, not autoregressive exact. One training seed per arm.

Configuration (K)	2K	4K	8K	16K	Retrieval (%)
MHA (32)	44.24/43.62	61.75/58.09	124.0/106.2	213.5/185.3	62.0/64.7
GQA-4 (32)	49.37/50.76	59.67/60.73	112.6/101.0	191.0/162.6	55.3/52.7
GQA-2 (32)	50.56/50.56	67.98/57.45	130.5/99.3	220.2/172.9	62.0/51.3
MLA-32 (16)	65.47/65.59	98.53/104.0	240.1/225.1	400.2/363.7	57.3/50.0
MLA-16 (8)	67.13/65.92	68.07/69.84	245.5/128.2	653.4/340.4	48.7/51.3

All five configurations improve long PPL, with the largest relative effect in the eight-pair MLA setting. The ordering across architectures is nonmonotone: KV sharing, latent compression, and rotary-pair count jointly define the setting. These results complement the three-seed 432M study by demonstrating allocation benefits across several attention architectures.

E SUPPORTING RESULTS

E.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified $\tau = 1.414$ setting. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree.

Evaluation corpus and aggregation. The archived evaluator reads the shared cached FineWeb-Edu stream `val_fineweb-edu_5000000.pt`, nominally five million GPT-2 tokens. For each method and seed it resets NumPy RandomState to 9999, iterates through lengths 8192, 16384, 20480, 24576, 28672, and 32768, and draws eight distinct start offsets per length. Every chunk scores all $L - 1$ next-token targets, with full-sequence cross-entropy and no tail restriction. Its per-model PPL is the exponential of the mean chunk NLL; the table then takes the arithmetic mean and sample standard deviation of the three per-seed PPLs. Thus the requested targets per model are $8(L - 1)$ at each length; starts are sampled without replacement, but overlapping windows are possible. The same offsets are reused across methods.

All methods use the same fixed FineWeb-Edu evaluation cache and paired chunk offsets. The archived result JSON retains every seed and length, allowing the reported means, seed variation and scaling comparisons to be recomputed.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 18: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond, showing that the allocation also composes with this wavelength-dependent extension.

E.2 ADDITIONAL MODEL COMPARISONS

The following comparisons provide protocol-level detail for the main results.

Table 19: **Table and amplitude separated on the Qwen-2.5-3B 36-row panel (%)**. Matching the coefficient across tables makes the table contrast comparable; the last row varies amplitude alone at a fixed table.

Table	Coefficient c in $g = 1 + c \ln s$	32K	128K
MrRoPE-Pro	0.1	87.22	78.13
BM	0.1	91.67	70.83
MrRoPE-Pro	0.074	98.33	75.35
BM	0.074	100.00	70.00
BM	$g = 1$	89.58	58.82

Table 20: **Matched six-task RULER comparisons (%)**. Each column averages six tasks ($s = 4$; short/long prompt counts per arm).

Model	Lengths	Rows	Short		Long	
			MrPro	BM	MrPro	BM
OLMo-1B	4/16K	24/48	37.85	81.81	2.78	51.32
Qwen-3B	32/128K	12/24	87.22	91.67	78.13	70.83
Qwen-7B	32/128K	6/12	83.33	80.00	84.44	71.11

Table 21: **Static extension across native and extended windows**. Qwen-0.5B RULER-13 macro (%); g is fixed amplitude.

Installed table	g	32K	64K
Native	1.0000	54.78	22.05
YaRN-2	1.0693	55.94	45.37
Normalized-index-2	1.0513	55.92	51.46

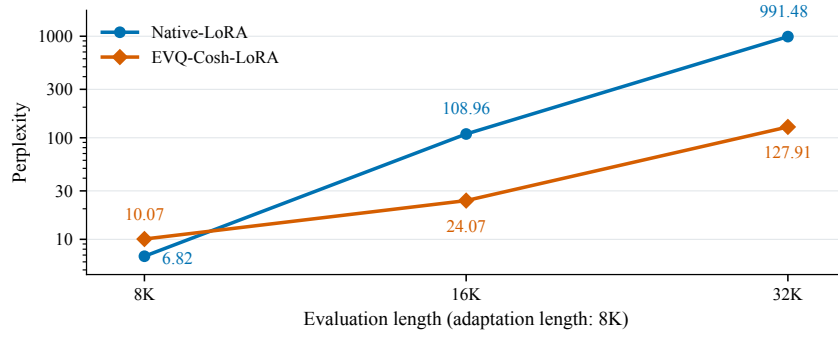


Figure 11: **Cosh allocation improves extrapolation under matched adaptation.** PPL is the exponential of equal-domain mean NLL.

E.3 LEARNING AND NATURAL-QA SUMMARIES

The 750M continuation starts from a common 2K Geo checkpoint and uses 500M tokens at 4K, including 3% passkey and 10% downstream mix. Its 8K task endpoint is greedy answer-token exact, not complete-response exact. The OLMo adaptation pair trains on physical inputs at most 4K with explicit target-range positions; evaluation uses real contiguous prompts. The Q/K-only RULER adaptation has Cosh-Native macro differences $-29.75/+29.61/+4.65$ points at 4/8/16K. These are learned protocols, not static-installation evidence. Small frozen Qwen 128K panels favor MrPro over BM (78.13/70.83% at 3B and 84.44/71.11% at 7B). Frozen Qwen-0.5B improves 64K RULER-13 over YaRN by 6.09 points (Appendix G.1). These older results retain their local protocols and do not substitute for the current clean comparison.

F MATURE-SCALE PROTOCOLS

F.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the standard $u_k=k/K$ grid with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, with explicit position IDs providing target-range relative phases. On 8K niah_single_1, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set.

Complete numeric answers after matched routing continuation. A separate follow-up to the routing parents applies the same 100-step query-gap continuation and 32-step answer-plus-EOS continuation to Native and EVQ-COSH (training seed 20260728). Every backward pass uses at most 4096 physical tokens; explicit position IDs expose target-range phases, with maximum observed position ID 16,257. Evaluation uses real contiguous prompts from the same numeric NIAH family and greedy decoding. The endpoint requires the complete expected answer string followed by observed terminal EOS, measured from generated token IDs.

Stage identities and supervision. The starting artifacts are the two 300-step counterfactual-routing parents (rank-64, alpha-128 Q/K/V/O LoRA). Their SHA-256 prefixes are 6570ab94aec6 (Native) and 95ceeb70117c (Cosh); the later stages continue each arm’s own parent. Table 24 specifies the two matched follow-ups. Both use fused AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$, zero weight decay, gradient clipping 1, BF16, and cosine decay to 0.1 of peak LR after linear warmup. Global

Table 22: **Allocation benefits across learning protocols.** Each block names its reference. MLA uses unanchored midpoint Cosh and 8K training; the 750M pair fixes endpoints. OLMo adaptation uses physical inputs $\leq 4K$ with explicit long-phase exposure and requires the full answer plus EOS.

Protocol	Metric	Reference	Cosh
<i>MLA learning: Geo reference, three seeds</i>			
50% budget	PPL at 16K	166.2	118.0
75% budget	PPL at 16K	146.2	101.6
100% budget	PPL at 16K	138.8	95.6
<i>750M continuation: Geo reference, one seed</i>			
After continuation	PPL at 4K	22.0	22.3
	PPL at 16K	45.1	24.4
	8K answer-token exact	0/40	31/40
<i>OLMo adaptation: Native reference, one matched pair</i>			
After phase exposure	4K complete + EOS	95/100	100/100
	8K complete + EOS	18/100	98/100
	16K complete + EOS	0/100	60/100

Table 23: **A concrete redistribution improves natural QA.** Frozen OLMo, static $s = 4$, whole-response token F1 (%). Both methods share the declared band, gain and inputs. The 631 inputs exceed the 4K native window; task means are equally weighted. Differences use unrounded scores. The paired, task-stratified 95% interval for the macro difference is $[1.32, 6.29]$ points.

Task	n	MrRoPE-Pro	BM	Difference
HotpotQA	166	28.63	33.54	+4.91
2WikiMQA	173	21.55	25.43	+3.88
Qasper	119	13.94	19.06	+5.11
NarrativeQA	61	9.51	11.48	+1.96
MultiFieldQA	112	34.49	37.72	+3.23
Task-equal mean	631	21.62	25.44	+3.82

batch 8 uses microbatch 4 and accumulation 2. The trainable Q/K/V/O adapters contain 16,777,216 parameters; the base checkpoint is frozen.

Table 24: **The matched continuation producing complete-answer outcomes.** Training seed 20260728. Each cycle contains two counterfactual-routing updates and one natural-text update. Counts include both variants of a routing pair.

Stage	Steps	Peak LR / warmup	Supervision and position policy
Query-gap	100	$5 \times 10^{-5} / 20$	Answer CE plus counterfactual margin; semantic query-block offsets through 16K.
Answer + EOS	32	$10^{-5} / 4$	Answer CE, immediate EOS CE (weight 1), same counterfactual margin; realized source-to-answer gap curriculum.

Routing examples contain eight key-value blocks and one query. The two variants replace the queried source value while preserving the surrounding context; the answer changes accordingly. The training set has 1024 paired examples, the separate calibration set 128, with seeds 20293725 and 20304725 and different query templates. The final stage reuses the raw training rows with rebuilt answer-plus-EOS labels, covering three answer tokens and EOS; padding and other prompt tokens carry no answer loss. The counterfactual term uses margin 1 and weight 0.5, comparing the correct and swapped answer where their tokens differ. Natural replay uses ordinary next-token loss. The 100-step stage contains 67 routing/33 natural updates; the final stage contains 22/10.

Position-ID construction. Let q be the semantic query-block start, a the answer start, h the end-exclusive source span, and n the active sequence length. Context positions before q remain contiguous.

Both counterfactual variants share the position array:

$$p_i = i + o \mathbf{1}\{i \geq q\}.$$

In the 100-step stage, each routing update uses four pair offsets: a low offset alternating between 0 and $[1, 4096]$, one from $[4097, 8192]$, and two from $[8193, 12289]$. Values traverse each band by a seed-derived coprime modular stride, with a deterministic per-update permutation. In the final stage, the stream instead specifies realized gap targets: the low target alternates between contiguous and $[4096, 8191]$, one target lies in $[8192, 12287]$, and two in $[12288, 16079]$. For a noncontiguous target g , set

$$o = \min\{g - (a - h), 12289, 16384 - \min(n, 4095)\}.$$

The implementation requires the resulting gap to remain in the target band and all active position IDs below 16384. Natural replay remains contiguous. The exact two deterministic schedules are included in `figs/verify_routing_schedule.py`; recomputation matches the run-record hashes for all 268 query offsets and all 88 final-stage gap targets. This procedure exposes target-range phases during training while keeping physical sequences at most 4096 tokens.

The evaluation uses separately generated, real contiguous numeric-NIAH prompts, 100 per length, with greedy decoding and complete-string-plus-EOS scoring. Training uses custom counterfactual query templates rather than the official evaluation template. The source receipts and full parent/final adapter hashes are included in `figs/routing_protocol_receipts.json` and `figs/recovered_asset_inputs.json`.

Table 25: Complete-answer-string plus terminal-EOS exact match after the matched routing follow-up, 100 inputs per length and one trained pair.

Length	Native	EVQ-COSH
4K	95/100	100/100
8K	18/100	98/100
16K	0/100	60/100

All final generations in both arms terminate with EOS. On the EVQ-COSH subsets whose source-query gap exceeds the original training range, exact match is 48/50 at 8K and 31/66 at 16K. These outputs belong to the additional continuation above, separately from the earlier first-number retrieval comparison.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled `Q/K metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 26 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals quantify evaluation-row variation for the trained pair.

Table 26: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on. The Llama rows use the separate RULER-family continuation: 516 steps, rank-64 Q/K/V/O LoRA, length 8192, seed 20420726. The 300-step LongAlpaca adaptation supplies the temporal-PPL and gold-block results, not these RULER scores.

Model / protocol	Arm / metric	1×	2×	4×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is +0.0724/+0.0381/−0.0437/−0.1351 at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K.

The EVQ-COSH branch uses a Hugging Face single-GPU loop; the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. This comparison includes the frequency-table change and the two trainer implementations.

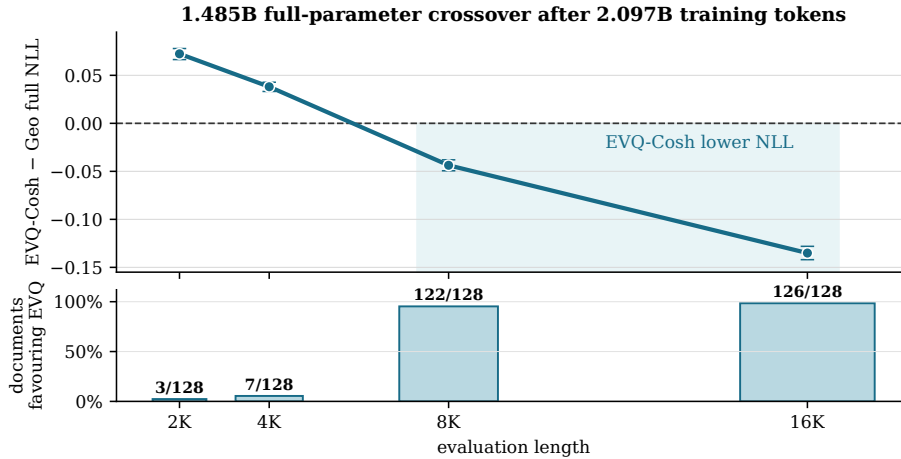


Figure 12: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

F.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 27 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The reference rows use the original Native table and the Transformers YaRN frequency

map. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $w_k = (1 - \tilde{U}_k)^2$ in the frequency blend of Eq. (54), so less unique pairs move farther.

Numerical construction of the fixed-support controls. The following specifies the derived arm independently of the later FullLagP2 construction. Use the checkpoint’s actual float32 native tensor promoted to float64, with $L = 4096$ for OLMo and 32768 for Qwen. Start from integer separations $d = 0, \dots, L - 1$ and weights $L - d$. Group consecutive separations in blocks of two, retain each block’s rounded weighted-centroid separation, and sum its weights; normalize the weights to unit mass. This gives 2048 and 16384 weighted support points, respectively.

Build X with columns $\sqrt{w_d} \cos(d\omega_k)$ and $\sqrt{w_d} \sin(d\omega_k)$ in float64. For each two-column block X_k , solve $B_k = \arg \min_B \|X_k - X_{-k}B\|_F$ with singular-value cutoff `rcond` = 10^{-10} , and compute

$$U_k = \text{clip} \left(\frac{\|X_k - X_{-k}B_k\|_F^2}{\|X_k\|_F^2}, 0, 1 \right).$$

If the marginal energy is zero, set $U_k = 0$. Min–max normalize the U_k values; if their span is zero, use the zero vector. Then $w_k = (1 - \tilde{U}_k)^2$ and $\omega'_k = \omega_k(1 - w_k) + \omega_k w_k/4$. Cast to float32, overwrite the first frequency with the exact native first entry and the last with the float32 native last entry divided by four, and require strict positivity and decreasing order. The resulting tensor SHA-256 prefixes are `a435d7544144` (OLMo) and `3512335e408c` (Qwen).

The uniform control is log-linear between the same endpoints. The coarse control enumerates all integer ramp bounds $0 \leq l < h < K$, minimizes mean squared distance to w_k in movement space, and breaks ties lexicographically in (l, h) . It uses the same frequency blend and endpoint overwrite. The selected bounds are (20, 22) for OLMo and (28, 31) for Qwen. FullLagP2 in Appendix G.9 instead uses every integer separation and installs a log-frequency shift; its measurements are a separate experiment.

Table 27: **Exact frozen mature-checkpoint RULER macro (%)**. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalued, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are [54.88, 64.80] points for OLMo and [0.25, 17.50] points for Qwen; the paired derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 28 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 28: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The OLMo deployment policy selects the exact Native table when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Table 29: Compact zero-training deployment endpoints on the frozen OLMo checkpoint, using the specified table, amplitude, and session routing.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$. The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Table 27 gives the corresponding allocation-only comparison.

F.3 LLaMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. Native uses the pretrained endpoint grid, and Cosh uses midpoint quantiles; the same adaptation recipe is applied to these two frequency tables. The frozen adaptation tensor is drawn from LongAlpaca-12k (Chen et al., 2023b): 7,476 training and 152 validation rows after length filtering, with effective batch 8 (microbatch 2, accumulation 4).

Temporal holdout and aggregation. Evaluation uses arXiv titles/abstracts, Federal Register titles/abstracts, and Stack Overflow question bodies selected by publication or creation timestamps between January 1 and July 12, 2026. These are separate from the LongAlpaca adaptation corpus. Collection was frozen before evaluation; each domain has eight disjoint-document 32,768-token packs formed by concatenation with EOS separators. Positions and causal attention continue across

document boundaries. The 8K and 16K conditions are exact prefixes of the same 32K packs. All ordinary text targets in each prefix are scored, excluding document-first tokens and inserted EOS separators. For domain d , let $\ell_d(L)$ and $n_d(L)$ be the summed token loss and valid-token count over its eight packs. The reported aggregate is

$$\text{NLL}(L) = \frac{1}{3} \sum_{d=1}^3 \frac{\ell_d(L)}{n_d(L)}, \quad \text{PPL}(L) = \exp(\text{NLL}(L)).$$

Table 30 gives token counts and the distribution of the 24 paired pack-level NLL differences. The accompanying `figs/llama_temporal_summary.json` retains each pack’s loss sum and count, its domain identity, and the frozen collection hash.

Table 30: Llama-3-8B temporal holdout: scored tokens and the 24 paired pack NLL differences (EVQ-COSH minus Native). Each length uses the same packs.

Prefix	Valid tokens/arm	Valid tokens/pack	Pack Δ NLL: min / median / max
8K	194,794	8,051–8,183	+0.193 / +0.432 / +0.613
16K	389,672	16,111–16,353	−2.069 / −1.436 / −1.278
32K	779,400	32,243–32,679	−2.319 / −2.074 / −1.754

The domain-macro EVQ-COSH-minus-Native NLL is +0.390 / −1.510 / −2.048 at 8/16/32K.

Position-preserving source intervention. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry. NLL pools the negative log probabilities of the 30 answer tokens (3 per case); deletion changes this mean by −0.0095 for Native and +1.5055 for EVQ-COSH. The EVQ-COSH change is positive in all ten cases. Case-level hit@16 ranges are 3.12–34.38% for Native and 62.50–68.75% for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

Generation and natural-QA outcomes of the same adapters. The same three-arm source collection also includes the unadapted base model. Its domain-macro prefix PPL at 8/16/32K is 7.948/150.545/1492.915, compared with 6.817/108.958/991.475 for Native-LoRA and 10.068/24.068/127.911 for Cosh-LoRA. This adds the adaptation baseline while retaining the 8K cost; the matched Native-LoRA comparison remains the control for generic adaptation. The portable three-arm prefix summaries retain their source collection identity.

The source interventions reveal how remote evidence affects answer probabilities. Task evaluation measures whether those probabilities produce the required output. In the seed-42 capability evaluation, Cosh passkey exact is 100/0/0% at 8/16/32K, and single-needle top-1 is 56.67/3.33/0%. The registered three-arm QA comparison gives task-macro F1 0.2309 for the base model, 0.2110 for Native-LoRA, and 0.1126 for Cosh-LoRA; the paired Cosh-minus-Native difference is −0.0984, with 95% bootstrap interval [−0.1297, −0.0697]. The deficit is concentrated in inputs of at most 8K; on the longer subset all arms are near the floor. Forced-gold inclusion changes NLL by −0.034, and the sparse readout pilot does not improve conversion. These report-backed outcomes accompany the probability and source-use measurements of the same 300-step adapters. They motivate distinguishing access to a distant source from successful task readout.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 26 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

G FROZEN EXPONENT-ADJUSTMENT PROTOCOLS

This section specifies the static profiles and evaluation panels used in §3.2 and the BM natural-QA comparison in §6. An amplitude g multiplies both the cosine and sine tables, so its contribution to a rotated Q/K logit is g^2 . Every table is installed before prefill and retained for the entire request. The accompanying `figs/recorded_runtime_identities.json` records checkpoint, tokenizer, and evaluator identities. The index/placement studies use RULER commit `c3f5e3b4f87f97e048793bb510a3a6b19a46bf3a`.

For the native-relative displacement in (2), the three installation laws are

$$d_k = \begin{cases} -\log(1 - w_k + w_k/s), & \text{frequency blend,} \\ m_k \log s, & \text{log-frequency shift,} \\ \sum_{j < k} \log \lambda_j, & \text{mixed-radix conversion.} \end{cases} \quad (54)$$

The blend weight w_k and log-shift profile m_k have different meanings; the first line is the exact conversion between them.

YaRN implementation coordinate. The index-linear YaRN baseline uses $r_k = \text{clip}((k - l)/(h - l), 0, 1)$ and $\omega'_k = \omega_k(1 - r_k + r_k/s)$, with the recorded correction indices and separate cosine/sine gain. This is the channel-index ramp in the author implementation (revision 995db5b) and its Transformers implementation (Peng et al., 2024). It is distinct from the historical smoothstep scaler in Appendix D and the MLA wavelength blend in Appendix E.1.

G.1 NORMALIZED-INDEX PROFILE

For a native table with K pairs, base b , and reference length L_{ref} , the normalized-index construction evaluates the profile on a 64-pair reference grid with the same b, L_{ref} :

$$\bar{\omega}_j = b^{-j/64}, \quad c_{64} = (1 - b^{-1/64})^{-1}, \quad \xi_j = \log \frac{L_{\text{ref}} \bar{\omega}_j}{2\pi c_{64}}, \quad (55)$$

$$\bar{m}_j = \text{clip} \left(\frac{\xi_H - \xi_j}{\xi_H - \xi_L}, 0, 1 \right), \quad \xi_H = 0.7382780681078285, \quad \xi_L = 0.366403835112904. \quad (56)$$

These two constants were fitted to the existing OLMo displacement profile and then frozen. Linear interpolation of $(j/63, \bar{m}_j)$ at $k/(K - 1)$ gives m_k on the target grid. The installed table is $\omega'_k = \omega_k^N s^{-m_k}$.

The Qwen-2.5-0.5B-Instruct experiment uses its native $K = 32$ pairs, $b = 10^6$, and original context length $L_{\text{ref}} = 32768$, with $s = 2$ and fixed amplitude $g = 1 + 0.074 \log 2 = 1.0512928914$. The YaRN-2 comparison uses the Transformers index-linear YaRN table and $g = 1 + 0.1 \log 2 = 1.0693147181$; Native uses $g = 1$.

The full RULER-13 confirmation uses seed 202609027, 20 examples per task and length, the same tokenizer and native chat template, and greedy task-budgeted generation. It contains 520 generations per arm across 32768 and 65536 tokens. Per-task scores are shown in Table 31. The macro averages the thirteen task means equally. Paired intervals use 10,000 bootstrap samples within task strata with seed 202609028. The 64K index-minus-YaRN difference is 6.0897 percentage points, with interval $[2.7627, 9.5835]$; at 32K it is -0.0256 points, with interval $[-3.2630, 3.2179]$.

Table 31: Qwen-0.5B RULER-13 per-task score (%), 20 examples per cell.

Task	Native	32K		Native	64K	
		Index	YaRN		Index	YaRN
Single-1	100.00	100.00	100.00	75.00	100.00	100.00
Single-2	100.00	100.00	100.00	15.00	100.00	75.00
Single-3	100.00	100.00	100.00	30.00	100.00	95.00
MultiKey-1	90.00	95.00	90.00	50.00	90.00	65.00
MultiKey-2	50.00	60.00	35.00	0.00	40.00	25.00
MultiKey-3	5.00	0.00	10.00	0.00	5.00	5.00
MultiValue	62.50	81.25	81.25	22.50	63.75	58.75
MultiQuery	62.50	80.00	77.50	22.50	65.00	50.00
Variable tracking	48.00	42.00	46.00	3.00	28.00	36.00
Common words	2.50	2.00	2.50	7.00	0.50	0.00
Frequent words	46.67	31.67	35.00	31.67	46.67	40.00
QA-1	20.00	10.00	30.00	5.00	15.00	20.00
QA-2	25.00	25.00	20.00	25.00	15.00	20.00

G.2 PLACEMENT ON ANOTHER FINITE PAIR GRID

The alternative direct-gap construction evaluates the same clipped profile using the target table’s local frequency spacing:

$$c_K = (1 - b^{-1/K})^{-1}, \quad \xi_k^{(K)} = \log \frac{L_{\text{ref}} \omega_k^N}{2\pi c_K}, \quad m_k^{\text{gap}} = \text{clip} \left(\frac{\xi_H - \xi_k^{(K)}}{\xi_H - \xi_L}, 0, 1 \right).$$

Both placements share the frozen ξ_H, ξ_L and amplitude coefficient 0.074. Within each comparison, endpoints, scale, amplitude, checkpoint, inputs, decoder, and scorer are matched. Table 32 reports two independent-input confirmations with 80 rows per task.

Qwen-0.5B uses the 32-pair, $s = 2$ setting above and data seed 202609026, with four tasks at each of 32K and 64K. These are separate inputs from the subsequent RULER-13 confirmation. Gemma-1.1-2B-Instruct uses $K = 128$, $b = 10,000$, a fixed operating reference of 4096, $s = 4$, and $g = 1 + 0.074 \log 4 = 1.1025857827$. Its confirmation uses data seed 202609028 at 16K. The task vector is NIAH single-1, multikey-2, multikey-3, and variable tracking. Gemma’s direct-gap scores are 100/88.75/20/82.50%; normalized-index scores are 100/88.75/40/87.25%. Both tables were frozen before these inputs were generated, and the earlier selection rows are excluded. This confirmation compares the two placements; the Native/YaRN baseline panel uses a different, earlier input set.

Table 32: Independent-input placement comparisons, 80 examples per task and four tasks per panel. Intervals are for index minus direct-gap, in percentage points: 97.5% for each Qwen length (Bonferroni over two lengths), 95% for Gemma.

Model	Length	Direct-gap (%)	Index (%)	Difference interval
Qwen-0.5B	32K	49.69	53.37	$[-0.25, +7.75]$
Qwen-0.5B	64K	46.62	46.13	$[-4.88, +3.81]$
Gemma-2B	16K	72.81	79.00	$[+2.81, +9.63]$

G.3 BOUNDARY-MATCHED INTERMEDIATE EXPONENTS

Let l be the last native frequency pair making more than 32 turns over the reference window and h the first making less than one turn. Write $N = h - l$ and $q = \text{clip}(k - l, 0, N)$. All profiles leave the fast band unchanged and divide the slow-band frequencies by s :

$$\omega'_k = \omega_k^N s^{-m_q}, \quad m_0 = 0, \quad m_N = 1. \quad (57)$$

With the same band boundaries, the three intermediate allocations are

$$m_q^{\text{Uni}} = \frac{q}{N}, \quad (58)$$

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad (59)$$

$$m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (60)$$

The Uni and Pro rules follow the uniform and progressive mixed-radix constructions of Tian et al. (2026). BM smooths the radix increments $\epsilon_q = m_q - m_{q-1}$ at both band boundaries.

To obtain BM, minimize $\sum_{q=0}^N (\epsilon_{q+1} - \epsilon_q)^2$ with $\epsilon_0 = \epsilon_{N+1} = 0$ and $\sum_{q=1}^N \epsilon_q = 1$. The discrete Euler equation gives

$$\epsilon_q = \frac{6q(N+1-q)}{N(N+1)(N+2)};$$

summing gives (60). The quadratic form is strictly convex with the endpoints fixed, so this solution is unique. Figure 13 shows the actual profile shapes for the tested transition widths.

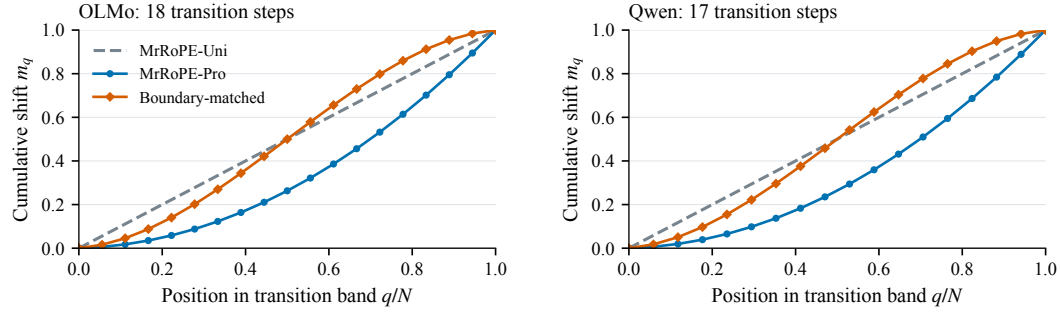


Figure 13: **Intermediate-band exponent displacements.** All three profiles have the same cumulative scale at the endpoints. BM redistributes the increments toward the middle of the transition band.

OLMo-2-0425-1B-Instruct uses $b = 500,000$, reference length 4096, $K = 64$, and $(l, h, N) = (14, 32, 18)$. Qwen-2.5-3B/7B-Instruct use $b = 10^6$, reference length 32768, $K = 64$, and $(l, h, N) = (23, 40, 17)$. The factor-four comparisons use $g = 1 + 0.1 \log 4 = 1.1386294361$ for both MrRoPE-Pro and BM, with FP32 frequency tables and BF16 model execution.

G.4 SIX-TASK MODEL COMPARISONS

The shared task types are NIAH single-2, multikey-2, multiquery, variable tracking, frequent words, and QA-1. OLMo’s confirmation uses seed 20260910 with four short and eight long examples per task, after a separate development panel with seed 20260909; BM’s formula was fixed before the confirmation. Its 4K/16K scores are listed in Table 20. MrRoPE-Uni and official YaRN were subsequently evaluated on these same inputs, scoring 76.88/32.12% and 54.38/6.94%. These are follow-up controls on the BM–MrPro confirmation panel, not new independent confirmations. The Qwen-3B panel has two 32K and four 128K examples per task. The 7B screen reuses the first one short and two long examples per task; this selection was fixed before its outputs were inspected. Table 33 reports every task mean.

All arms use the corresponding native chat template and the same task-specific generation caps, score the entire saved response with the RULER task metric, and retain generated token IDs and EOS flags. On Qwen-7B, a tokenwise MLP chunk size of 4096 makes 128K evaluation fit in memory; both arms share that path while retaining full attention.

Table 33: All six-task BM/MrRoPE-Pro comparisons. Values are official RULER task scores (%); model-specific sample counts are given in Table 20.

Model	Task	Short		Long	
		MrPro	BM	MrPro	BM
OLMo-1B	Single-2	75.00	100.00	0.00	100.00
	MultiKey-2	25.00	100.00	0.00	37.50
	MultiQuery	18.75	87.50	0.00	43.75
	Variable tracking	25.00	70.00	0.00	10.00
	Frequent words	58.33	58.33	16.67	41.67
	QA-1	25.00	75.00	0.00	75.00
Qwen-3B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	75.00	50.00
	MultiQuery	100.00	100.00	93.75	100.00
	Variable tracking	90.00	100.00	75.00	75.00
	Frequent words	83.33	100.00	75.00	75.00
	QA-1	50.00	50.00	50.00	25.00
Qwen-7B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	100.00	100.00
	MultiQuery	100.00	100.00	100.00	100.00
	Variable tracking	100.00	80.00	90.00	60.00
	Frequent words	100.00	100.00	66.67	66.67
	QA-1	0.00	0.00	50.00	0.00

G.5 NATURAL QUESTION ANSWERING

The natural comparison reuses frozen, source-verified LongBench prompts (Bai et al., 2024). Inputs are untruncated, retain their source questions and references, and fit the 16K input-plus-generation budget. Selection follows the pre-existing source-hash order; no generated answer participates in selection. The complete eligible pool contains 778 questions, of which 631 have more than 4096 input tokens. Both methods retain the same factor-four table at every length.

Greedy generation uses a 32-token cap for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. The metric is token F1 on the complete saved generated response against the task references. Table 34 gives the long-input and original-window-length strata, including EOS counts. The long-input macro averages the five task means; the shorter-input macro averages its four available tasks. The respective BM-minus-MrPro changes are +3.8194 and +2.0028 points, with paired within-task bootstrap intervals [1.3161, 6.2920] and [-3.5618, 7.4742]. The MultiFieldQA long-input row includes 40 capped responses in each arm; these remain in the F1 calculation.

Table 34: Natural QA under static $s = 4$. Token F1 uses the whole generated response; EOS counts are recorded separately.

Input length	Task	n	MrPro F1 (%)	BM F1 (%)	MrPro EOS	BM EOS
4K–16K	HotpotQA	166	28.63	33.54	160	163
	2WikiMQA	173	21.55	25.43	169	171
	Qasper	119	13.94	19.06	119	119
	NarrativeQA	61	9.51	11.48	61	61
	MultiFieldQA	112	34.49	37.72	72	72
$\leq 4K$	HotpotQA	5	58.33	51.67	5	5
	2WikiMQA	26	34.62	44.98	26	26
	Qasper	78	26.16	28.14	78	78
	MultiFieldQA	38	50.94	53.27	35	33

G.6 COMPACT DESCRIPTIONS OF A USEFUL ALLOCATION

An ordered 64-slot movement profile can be approximated by a compact law without retaining every local feature. Define $c_k = L\omega_k/(2\pi)$, $c_{\text{orth}} = (1 - b^{-1/K})^{-1}$, and $\xi_k = \ln(c_k/c_{\text{orth}})$. The adjacent-frequency separation is exactly $L(\omega_k - \omega_{k+1})/(2\pi) = e^{\xi_k}$. A clipped-affine law fitted to the frozen OLMo movement vector is

$$G_4(\xi) = \text{clip}\left(\frac{\xi_H - \xi}{\xi_H - \xi_L}, 0, 1\right), \quad \omega'_k = \omega_k 4^{-G_4(\xi_k)}.$$

The fitted boundaries are $\xi_H = 0.7382780681$ and $\xi_L = 0.3664038351$. The two parameters were frozen before examining Qwen geometry or model results. Both checkpoints use the fixed cosine/sine gain $1 + 0.074 \ln 4$. The fit has movement MAE 0.001223 on OLMo. The original OLMo-to-Qwen transport interpolates the full profile in $u = e^\xi$; C2 evaluates the frozen fitted law directly in ξ .

Table 35: **A compact profile preserves long-context behavior.** Official task macro (%). OLMo uses RULER-13; Qwen uses four core tasks. The tables were frozen before the long-length evaluation; Qwen 32K was collected afterward.

Model	Endpoint	Full 64-point profile	C2
OLMo	8K	66.71	66.84
OLMo	16K	49.86	50.24
Qwen	32K	70.00	71.25
Qwen	64K	67.25	67.75
Qwen	128K	54.50	57.25

The long-end results show that detailed 64-point structure is not required to preserve these measured behaviors. At the native window, OLMo’s PPL retention is 0.870971 and task retention 0.904932; Qwen native-task macro is 82.00% versus 71.25% for C2. The compact approximation preserves long-context scores while changing native-window behavior. The original profiles, C2, and the later normalized-index placements retain distinct tensor identities.

G.7 AN INTERIOR-PROFILE CONTROL AT FIXED TOTAL DISPLACEMENT

For a fixed band of width n and unit increment mass, define $m_q = \sum_{r=1}^q \epsilon_r$ and centroid $\mu = \sum_{r=1}^n r\epsilon_r$. Reordering the finite sum gives

$$\sum_{q=1}^{n-1} m_q = \sum_{r=1}^n (n-r)\epsilon_r = n - \mu. \quad (61)$$

At a common scale $s > 1$ with identical outer bands, equal total log displacement therefore fixes μ ; it is not an additional independent control. Remaining contrasts concern higher-order increment shape.

For $K = 64$, let $k = 1, \dots, 18$, $p_k = 6k(19-k)/(18 \cdot 19 \cdot 20)$, and $v_k = k - 19/2$. Two increment profiles are

$$\epsilon_k^A = p_k \left(1 - \frac{10}{119}v_k\right), \quad \epsilon_k^B = p_k \left(1 - \frac{10}{119}v_k + \frac{35}{1496} \left(v_k^2 - \frac{357}{20}\right)\right).$$

Both use $m_j = 0$ through slot 14, cumulative increments on slots 15–32, and $m_j = 1$ thereafter. They have unit increment mass, identical sampled endpoints, and $\sum_j m_j = 42$; their equal centroids follow from Eq. (61). Installed frequencies are $\omega'_j = \omega_j 4^{-m_j}$ with fixed gain $1 + 0.1 \ln 4$ and terminal shift $m = 1$. This comparison tests higher-order increment shape at equal total displacement.

On 350 shared 16K OLMo development inputs, scores are 43.6714% and 54.4000%. The paired difference is +10.7286 points, with 75 wins, 26 losses, and 249 ties; the paired standardized mean is $t = 5.47$. A separate 16-document continuous-text evaluation reports NLL 2.9417 versus 2.8309. These observations distinguish two internal profiles that share their total displacement. The task panel was used in profile exploration. The numerical construction, row-score aggregation, and common-total checks are reproduced by `figs/verify_recovered_assets.py`.

G.8 FOUR METHODS ON THE SAME OLMo TASK PANEL

The four-method table uses the 24 short and 48 long prompts of the seed-20260910 BM–MrPro confirmation panel (Appendix G.4). Uni and YaRN were added as follow-up controls on those same inputs. All four methods use $s = 4$ and gain $1 + 0.1 \ln 4$. MrRoPE profiles are local formula implementations; official YaRN denotes the channel-index ramp implementation. Table 36 is recomputed from the recorded per-task aggregates, with matched BM/control comparisons recorded on the shared row identities.

Table 36: **Intermediate-profile comparison on the OLMo six-task panel.** Official task macro in percent, including partial-credit tasks.

Method	4K	16K
MrRoPE-Pro(4)	37.85	2.78
Official YaRN(4)	54.38	6.94
MrRoPE-Uni(4)	76.88	32.12
BM(4)	81.81	51.32

An equal-total-displacement shape control. For $q = 1, \dots, n - 1$, summing the cumulative profiles gives

$$\sum_q m_q^{\text{BM}} = \sum_q m_q^{\text{Uni}} = \frac{n-1}{2}, \quad \sum_q m_q^{\text{Pro}} = \frac{n-1}{3}.$$

The outer bands agree, so BM and Uni have identical full-table displacement, whereas BM exceeds Pro by $(n-1) \log s/6$. On the actual $K = 64$, $[l, h] = [14, 32]$ grid, full-table $\sum_k m_k$ is 40.5 for BM/Uni and $37\frac{2}{3}$ for Pro. The 16K BM–Uni difference of 19.20 points therefore compares different shapes at equal total displacement. This follow-up comparison tests shape at fixed total displacement; natural QA and C42 use their own panels.

Appendix G.5 evaluates BM and MrRoPE-Pro on natural QA inputs.

G.9 FULL-LAG POSITIONAL GEOMETRY AS A CANDIDATE CONSTRUCTION

FullLagP2 constructs the native sine–cosine column pairs across all separations $d = 0, \dots, 32767$ with causal weights proportional to $32768 - d$. For each pair it computes its least-squares residual fraction relative to the other pairs (cutoff 10^{-10}), min–max normalizes these fractions to $\bar{u}_j$, and sets $m_j = (1 - \bar{u}_j)^2$. The resulting full table uses $s = 4$ and gain $1 + 0.074 \ln 4$. It depends on positional geometry, with no model-weight or activation fitting.

Table 37: **Recorded FullLagP2 task outcomes on Qwen-1.5B.** Eight prompts/task/length, official task score in percent. The matched-gain MrPro arm uses the same gain as FullLagP2.

Length	Task	MrPro	MrPro, matched gain	FullLagP2
64K	MultiKey-2	12.50	25.00	37.50
	Variable tracking	82.50	77.50	87.50
	Frequent words	45.83	45.83	70.83
128K	MultiKey-2	0.00	–	0.00
	Variable tracking	72.50	–	85.00
	Frequent words	50.00	–	50.00

This is a positive small-panel example of converting full-pair geometry into a useful allocation. With the same full array on Qwen-3B at 64K (four prompts/task), FullLagP2/MrPro scores are 75/50% for MultiKey-2, 90/95% for variable tracking, and 66.67/75% for frequent words. The decoder uses repetition penalty 1.1; these outputs remain separate from later screening panels. The paired score fractions and checkpoint identities are preserved in `figs/recovered_asset_inputs.json`. The construction’s geometric residual is a design statistic, and the task measurements establish where it is useful.

H TAILSPLINE: CONSTRUCTION AND FROZEN EVALUATION

H.1 FINITE-GRID PROOF

Let $D\epsilon = (\epsilon_{q+1} - \epsilon_q)_{q=1}^{n-1}$ and $H = D^\top D + e_n e_n^\top$. A zero value of $\epsilon^\top H \epsilon$ requires constant increments and a zero terminal increment, hence $\epsilon = 0$. Thus H is positive definite. For $n > 1$, the proposed $\epsilon_q = c(n+q)(n-q+1)$, with $c = 3/[n(n+1)(2n+1)]$, satisfies

$$\epsilon_1 - \epsilon_2 = 2c, \quad 2\epsilon_q - \epsilon_{q-1} - \epsilon_{q+1} = 2c, \quad 2\epsilon_n - \epsilon_{n-1} = 2c.$$

Therefore $H\epsilon = 2c\mathbf{1}$, the equality-constrained stationarity equation. The increments are positive and sum to one; summing them yields Eq. (8). The objective minimum is $2c = 6/[n(n+1)(2n+1)]$. For $n = 1$, the constraint gives $\epsilon_1 = 1$, agreeing with the formulas. Positivity makes additional nonnegativity constraints inactive.

Adding ϵ_1^2 yields the symmetric objective, with unique solution

$$\epsilon_q^{\text{BM}} = \frac{6q(n-q+1)}{n(n+1)(n+2)}, \quad m_q^{\text{BM}} = \frac{q(q+1)(3n+2-2q)}{n(n+1)(n+2)}. \quad (62)$$

Both profiles are monotone with $m_0 = 0, m_n = 1$. The portable CPU check independently solves the constrained tridiagonal system and verifies each closed form.

What the objective smooths. On the geometric native grid, Eq. (6) makes each change between neighboring log gaps equal to $(\epsilon_{q+1} - \epsilon_q) \log s$. The terminal change is $-\epsilon_n \log s$ because the fully interpolated tail has constant displacement. Thus $(\log s)^2$ times Eq. (7) is exactly the squared log-gap variation within the transition and at its low-frequency junction. The high-frequency junction is deliberately unpenalized.

At the terminal gap,

$$\epsilon_n^{\text{Pro}} = \frac{2}{n+1}, \quad \epsilon_n^{\text{TS}} = \frac{6}{(n+1)(2n+1)}, \quad \frac{\epsilon_n^{\text{TS}}}{\epsilon_n^{\text{Pro}}} = \frac{3}{2n+1}.$$

The opposite entry behavior is also explicit: $\epsilon_1^{\text{Pro}} = 2/[n(n+1)]$, whereas $\epsilon_1^{\text{TS}} = 3/(2n+1)$. TailSpline trades a larger entry jump for a smaller terminal jump. A useful boundary prior need not minimize both. In the continuum limit, $m^{\text{TS}}(u) = (3u - u^3)/2$ solves $\min \int_0^1 (m''(u))^2 du$ under $m(0) = 0, m(1) = 1, m'(1) = 0$; the natural entry condition is $m''(0) = 0$. The finite-grid formula is used in every model experiment, not this limiting polynomial.

H.2 AN EXACT EQUAL-DISPLACEMENT CONTROL

Use local indices $q = 0, \dots, n$. Write $T_q = m_q^{\text{TS}}, B_q = m_q^{\text{BM}}, U_q = q/n$, and $P_q = q(q+1)/[n(n+1)]$. Define the front-loaded profile $F_q = 2U_q - P_q$ and the control

$$C_q = (1 - w_n)U_q + w_n F_q, \quad w_n = \frac{3n}{2(2n+1)}. \quad (63)$$

Substitution of the finite-grid formulas gives $T = (1 - w_n)B + w_n F$. Subtracting C gives

$$m^{\text{TS}} - m^C = (1 - w_n)(m^{\text{BM}} - m^{\text{Uni}}). \quad (64)$$

In local coordinates, the explicit difference is

$$T_q - C_q = (1 - w_n)(B_q - U_q) = \frac{q(n-q)(2q-n)}{2n(n+1)(2n+1)}. \quad (65)$$

This cumulative-profile difference is antisymmetric under $q \mapsto n - q$, vanishes at both endpoints, and sums to zero. Since U and F have positive increments and $0 < w_n < 1$, C is monotone with $C_0 = 0, C_n = 1$. Extending both T and C by the same constant outer bands gives identical endpoints and total log displacement at a common scale. Their equal increment centroids follow from Eq. (61), rather than imposing a further constraint. For $n = 1, 2$, $B = U$ and $T = C$; for wider bands the contrast tests higher-order shape.

The reported TailSpline–MrPro results below change total displacement and shape together. The completed T – C diagnostic in Appendix H.8 tests the zero-total correction above under the recorded cross-batch runtimes. Its primary interval spans both signs. The algebra specifies the contrast, while the runtime-qualified result supplies its observed task response. The independent CPU check in `figs/allocation_design.py` verifies the finite-sum and profile identities with exact rational arithmetic.

H.3 PROTOCOL AND COMPLETE LENGTH RESPONSES

Frozen Llama-3-8B-Instruct has native window 8192. Both arms use exact finite-grid profiles, $s = 4$, canonical 32/1-turn band $[18, 35]$, and cosine/sine gain $1 + 0.1 \ln 4$. Each table is installed before prefill and shared across layers, requests and evaluation lengths. Construction uses no weights, activations or calibration outputs.

The depth-balanced RULER-13 panel contains 10 prompts per task at 8/16/32K: 390 paired inputs per arm. NIAH and passkey are subsets of this panel. PPL uses 32 Proof-Pile-test and 14 PG-19-test documents at each length: 138 rows per arm (Azerbaiyev et al., 2022; Rae et al., 2019). Both compared methods retain their $s = 4$ table even at the native input length.

Normalized trapezoidal log-length AUC weights the three endpoints by $1/4, 1/2, 1/4$. For PPL it integrates PPL over log length, not log PPL. At each length, PPL is the exponential of total whole-sequence loss divided by total scored tokens across the 46 documents. The paired document bootstrap resamples document identities jointly across lengths, recomputing token-weighted NLL, PPL and AUC for each draw. Task aggregates weight tasks equally. The recorded paired intervals characterize the tested inputs and task families for this frozen checkpoint.

Table 38: Recorded Llama responses. NIAH uses recorded paired differences; absolute scores are not reconstructed.

Endpoint	8K	16K	32K
RULER-13 MrPro (%)	87.5000	80.0897	54.7308
RULER-13 TailSpline (%)	89.7436	84.0513	57.3590
NIAH TailSpline–MrPro (pp)	−0.3125	+5.9375	+3.4375
PPL MrPro	5.286429	4.518111	4.161831
PPL TailSpline	5.295283	4.530318	4.110622

Family AUC differences are retrieval +3.750 points, tracking +6.500, aggregation approximately +3.792, and QA −1.250. Multikey-2 improves by 22.5 points; QA-1 and multikey-1 each decrease by 5.0. Out of 390 inputs, TailSpline and MrPro respectively have 30 and 37 empty outputs, 388 and 384 EOS-terminated outputs, and 2 and 6 budget-limited outputs. Task scores include partial credit.

The portable input `figs/allocation_value_inputs.json` records the aggregates and source identities. The packaged constructor computes the boundaries and FP32 table from the public native grid, reference length and scale. TailSpline and MrPro share their outer bands, endpoints and gain, so their matched comparison tests interior allocation. Their total log-frequency displacement also differs; at fixed support this is a statistic of z . The equal-total controls in Appendix G.8 further isolate shape at fixed displacement.

H.4 PROSPECTIVE OLMo CONFIRMATION

The OLMo-2-0425-1B-Instruct comparison uses original window 4096, canonical band $[14, 32]$, $s = 4$ and the same gain as Llama. Generation batch size is four. The methods and data were frozen before reading outputs from this experiment; historical BM/front-loaded work on OLMo had informed the direction. Each model has its own tokenized panel and within-model paired comparison.

Table 39: Complete OLMo length responses under the frozen two-arm contract.

Endpoint	4K	8K	16K
RULER-13 MrPro (%)	32.5128	14.8462	7.2436
RULER-13 TailSpline (%)	72.6667	72.5513	48.6154
NIAH MrPro (%)	33.1250	13.1250	7.1875
NIAH TailSpline (%)	88.7500	87.8125	60.0000
PPL MrPro	13.795244	14.356459	12.827048
PPL TailSpline	11.152472	10.117407	8.536196

Full-13 AUC difference is 0.492340 with paired 95% interval [0.453365, 0.530449]; NIAH difference 0.644531 has interval [0.592188, 0.694531]; PPL difference -3.852932 has interval $[-4.775042, -3.125459]$. Passkey AUC is 0.950 versus 0.375. All 13 task differences are positive, ranging from $+0.0325$ for common words to $+0.875$ for single-3. Family differences are retrieval $+0.64453$, tracking $+0.37000$, aggregation approximately $+0.16208$, and QA approximately $+0.27563$.

Neither arm has empty outputs. TailSpline and MrPro terminate with EOS on 257 and 248 of 390 inputs, respectively, and hit the cap on 133 and 142 (34.1% and 36.4%). PPL uses the same document sources as Llama with OLMo-tokenized prefixes. Report and original stream identities accompany the portable aggregates.

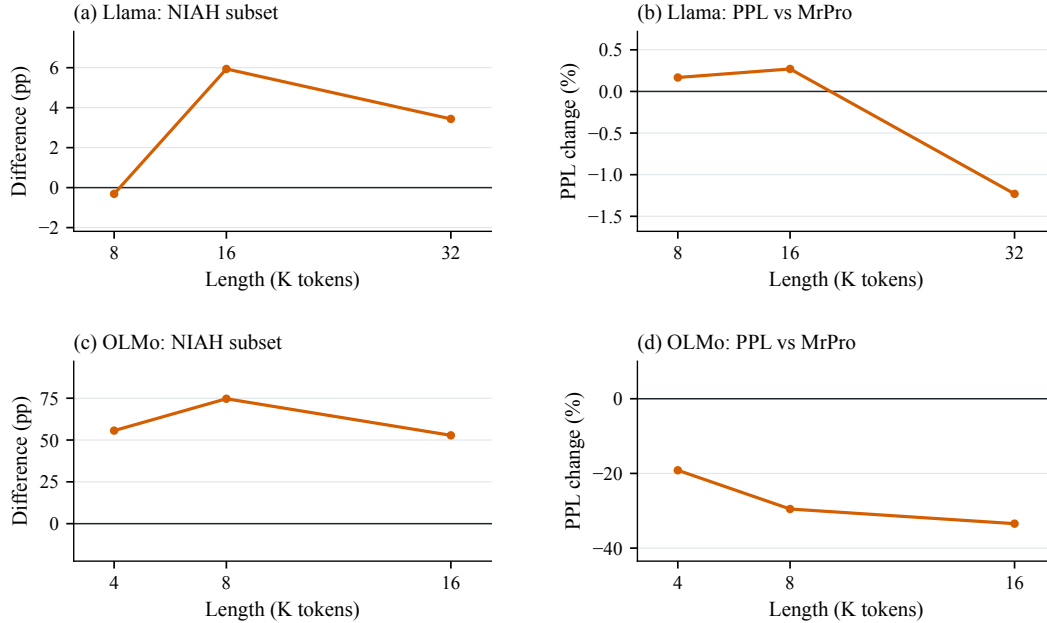


Figure 14: **NIAH and language-modeling responses.** NIAH is a subset of each model’s depth-balanced RULER-13 panel. Both columns show TailSpline relative to MrPro; PPL changes use each length’s absolute PPL. Llama’s 8/16K PPL slightly increases, while OLMo improves at every measured length. The panels have separate vertical scales.

H.5 EVALUATION IMPLEMENTATION

Generation is greedy, with repetition penalty 1, no repeat-ngram restriction and the model’s EOS IDs. Llama uses batch size one with cached 8192-token prefill chunks; OLMo uses batches of at most four identical-length inputs. The static FP32 frequency table is installed after model loading and its installed values are checked against the receipt. NIAH point-depth tasks use two prompts at each target depth 10/30/50/70/90% per length. Other tasks keep their recorded task-specific output budgets.

Input construction. The generator pins NVIDIA RULER revision c3f5e3b. Each NIAH task-length cell selects 10 rows from 64 candidates using only evidence positions and row identifiers. For single-evidence tasks, the five target depths above each receive two prompts; selection minimizes absolute depth error with tolerance 0.10. Multi-evidence tasks alternate dispersed targets (0.10, 0.35, 0.65, 0.90) and middle-cluster targets (0.35, 0.45, 0.55, 0.65), with mean absolute tolerance 0.18. Depths use token positions divided by the context cap, after padding. Task scores are not used in selection.

The filler unit is Irrelevant archive padding record; it does not answer the question. followed by a newline, tokenized without special tokens. Its token sequence is repeated and truncated to the required padding count, then inserted immediately after the user header, before the original task template, until the input occupies cap minus 128 tokens. The original context, question and references are preserved. In the pinned aggregation templates, the numbered list or coded text follows the task instruction; filler is placed before that instruction. The task-specific generation budgets remain 128 (NIAH), 30 (tracking), 120 (common words), 50 (frequent words) and 32 (QA). The row records store padding-token counts and achieved depth errors.

The Llama and OLMo preparation seeds are 20260924 and 20260925; each cell adds $300000 + 10000i + 100j$ for length index i and task index j . QA uses question-index blocks starting at 5000, 5010 and 5020 for the three lengths. These rules specify sample selection within each model; the two model panels have separate tokenizations. Both exact tables were fixed before their corresponding comparison; historical Llama and OLMo experiments informed the design direction.

Natural-text sources. Proof-Pile uses the original hoskinson-center/proof-pile test archive, selecting the first 32 arXiv records with at least 300000 characters and retaining up to the first million characters. PG-19 uses individual test-book prefixes. The frozen 46-document source manifest records the selected 32 Proof-Pile and 14 PG-19 identities; each tokenizer takes a prefix of maximum evaluation length plus one token. Documents are scored separately rather than concatenated.

Full-13 intervals use 20,000 paired resamples within each task-length cell, retaining fixed task weights; NIAH intervals aggregate the corresponding retrieval tasks. The report treats different length cells separately for this panel. PPL intervals instead resample the 46 document identities jointly across all lengths, also using 20,000 draws. The two uncertainty calculations therefore preserve different experimental units. These are conditional evaluation intervals, not uncertainty across training seeds or model families.

H.6 THE FIXED T/C EXCHANGE AND TASK RESPONSE

The profile difference in Eq. (65) satisfies $\delta_{n-q} = -\delta_q$, $\sum_q \delta_q = 0$ and $\delta_q < 0$ for $0 < q < n/2$. For a continuously differentiable utility $\mathcal{V}$ on a neighborhood of the C-to-T path, define

$$G_q = \int_0^1 \partial_{m_q} \mathcal{V}(C + t\delta) dt, \quad \mathcal{V}(T) - \mathcal{V}(C) = \sum_{q < n/2} (-\delta_q)(G_{n-q} - G_q). \quad (66)$$

The fundamental theorem of calculus gives the path integral; pairing indices gives the second expression. The hypothesis is that later-half slowdown has higher marginal task value for this fixed exchange. A sequence log-likelihood can be differentiable; greedy accuracy and F1 do not satisfy this premise. The G_q identity is explanatory, not an independent predictor or an optimization procedure. This exposition was clarified after the E1 outputs were available; it does not retroactively preregister a new prediction. No equivalence margin was preregistered for the current comparison.

From logits to value readout. For fixed visible keys and current-layer content Q/K/V, let η_j be the finite score change caused by switching tables. Direct softmax algebra yields

$$p'_j = \frac{p_j e^{\eta_j}}{\mathbb{E}_p e^{\eta}}, \quad \log \frac{p'_j/p'_r}{p_j/p_r} = \eta_j - \eta_r, \quad o' - o = \frac{\text{Cov}_p(v, e^{\eta})}{\mathbb{E}_p e^{\eta}}. \quad (67)$$

A common logit shift has no effect. Reweighting keys only helps an output when it changes the useful value mixture. A scalar gain absorbs a score change exactly only when the target scores equal a

common positive multiple of the original scores plus a row constant; a shared gain requires the same multiple across queries. These are standard identities, related to LeRoPE’s frequency–attention–value gradient chain (Karypis et al., 2026). They do not hold the entire Transformer trajectory fixed after a table swap. A mechanism claim requires a separate output intervention with runtime-matched no-op and shuffled controls; no such completed result is claimed here.

H.7 CLEAN SOURCE-ORDER RULER CONFIRMATION

The clean panel uses RULER upstream revision `c3f5e3b4f87f` (full revision in the data receipt), source-order selection, 200 rows per task and a 32,768-token cap. Actual input lengths are 28,270–32,606 tokens. It uses neither content padding nor depth balancing; it is distinct from the classic panel and smaller than upstream’s 500-per-task default. Each completed arm has 2,600 unique prompt hashes. Both run contracts use batch 1, no left padding, and prefill chunks of 8192; they differ only in the arm label and static table. Scores were recomputed from all stored answers using the recorded scorer. The single-length endpoint is the task-equal score at that length, with no division by a zero log-length span.

Selection and table reuse. The clean preparation takes the first 200 upstream source rows per task in file order, appending the fixed additional QA-1 source fragment when needed to reach that count. It prepares the shared input IDs before running either arm and reads no generated model answers or scores. The runner reuses the classic static table files; the retained TailSpline contracts have identical frequency arrays, gain and construction metadata. The change from classic to clean is in input selection and padding policy, with no new allocation fit. Full-13 task-equal score at 32K is the endpoint for this separate source-order panel.

Table 40: Clean RULER-200: all task means and output health. Scores include capped and empty responses.

Task	TailSpline (%)	MrPro (%)	Difference (pp)
cwe	1.65	0.30	+1.35
fwe	61.83	39.67	+22.17
niah_multikey_1	84.50	62.50	+22.00
niah_multikey_2	47.00	9.00	+38.00
niah_multikey_3	24.00	2.50	+21.50
niah_multiquery	95.25	94.88	+0.38
niah_multivalue	93.12	97.62	-4.50
niah_single_1	100.00	97.00	+3.00
niah_single_2	99.50	99.00	+0.50
niah_single_3	100.00	94.00	+6.00
qa_1	64.00	55.50	+8.50
qa_2	37.00	27.50	+9.50
vt	79.60	55.60	+24.00
Task-equal mean	68.27	56.54	+11.72
EOS count	2573	2517	—
Cap-hit count	27	83	—
Empty count	240	310	—
Generated tokens	54330	58894	—

The paired within-task bootstrap uses 20,000 draws, seed 20260923, with tasks fixed. The T-P macro interval is [10.32, 13.11] points. Intervals condition on the selected source records, not independent documents from the entire QA corpus. The portable score-only pairs retain source-record and prompt hashes for checking the point estimand. An independent bootstrap with a different frozen seed agrees within Monte Carlo variation.

Family differences are retrieval +10.86, tracking +24.00, aggregation +11.76 and QA +9.00 points; their separate 95% intervals are [9.11, 12.61], [20.60, 27.40], [8.80, 14.77] and [4.25, 13.75]. These are descriptive decompositions, not four independent confirmations or multiplicity-adjusted family claims. Leaving out any single task leaves a positive macro difference, from 9.53 to 13.07 points. The one negative task, `niah_multivalue`, remains in the primary score. EOS, cap-hit and empty

counts in Table 40 include every response; no answer was excluded for termination behavior. This result supports TailSpline over MrPro on this clean panel; it does not yet compare YaRN or establish natural-QA transfer.

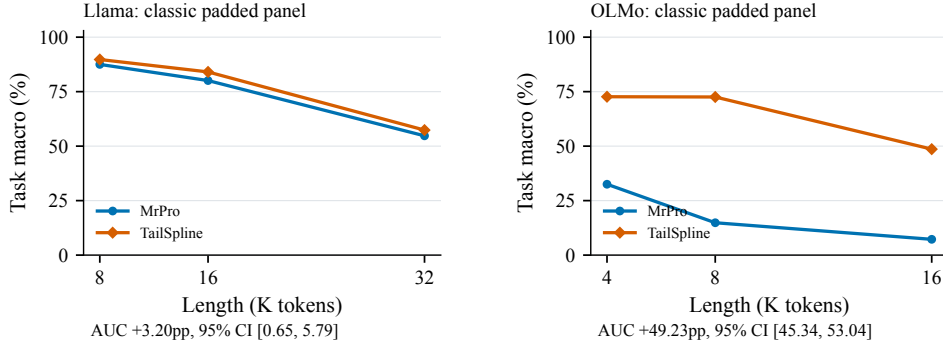


Figure 15: **Classic length responses on Llama and OLMo.** Each model uses its recorded depth-selected, prefix-padded input protocol. Curves and log-length AUC intervals complement the clean length panels in the main text.

H.8 CURRENT EQUAL-DOSE AND ORIGINAL-NATIVE CONTROLS

The completed E1 classic T/C comparison has 390 generations and 138 LM rows per arm. Raw task scores reproduce Full-13 AUC 78.8013% for T and 79.2115% for C. The reported T-C difference is -0.4103 points, interval $[-2.6282, 1.8205]$; the NIAH subset differs by $+2.0313$ points and PPL AUC by -0.000616 . These secondary endpoints do not replace Full-13. T/C have 388/380 EOS, 2/10 cap-hit and 30/22 empty outputs. Their installed $\sum m$ differs by -7.65×10^{-9} from float32 reconstruction; this does not change the exact same-dose construction.

Generation runtime differs: T uses batch 1, C batch 2 and reordered batches. Their legacy contracts also lack complete revision/backend identity fields. E0 replays one T row per task and length (39 rows) at batch 2: no official score changes, but six token sequences and decoded outputs change. Thus E1 remains a cross-runtime diagnostic. Neither E0 nor the interval permits an equivalence claim; a complete matched C arm and explicit runtime provenance would be needed for a stronger residual-shape attribution. The original E1 interval is a paired task-length-row analysis; it does not establish independent source blueprints across lengths or tail-only mediation.

The original-Native LM reference uses original frequencies and gain on 46 matched 8K documents, 376,832 scored targets. Whole-prefix token-pooled PPL is 5.275919 (Native), 5.295283 (T), and 5.286429 (MrPro). Corresponding NLL is 1.663153/1.666816/1.665143. T-Native PPL is $+0.019364$ (document-bootstrap 95% interval $[0.006602, 0.034861]$); MrPro-Native is $+0.010510$ ($[-0.000722, 0.024591]$). This whole-prefix LM reference is distinct from the classic PPL-AUC aggregation. It quantifies native-language-model cost and does not measure native generated-task retention.

The completed original-Native task reference uses the classic 8K slice (130 paired inputs), with task-equal macro 91.8846% (Native), 89.7436% (T), and 87.5000% (MrPro). Reversing the reported Native-T contrast gives T-Native -2.1410 points and interval $[-6.1410, 1.9231]$. The point difference is a 2.33% relative native-task trade-off. The interval is reported as paired uncertainty; no noninferiority margin was specified. This task comparison and the whole-prefix LM comparison quantify different costs.

H.9 TAILSPLINE ON THE FROZEN NATURAL-QA POOL

The Natural-QA631 V2 pool preserves the previously defined 631 natural questions from HotpotQA, 2WikiMQA, Qasper, NarrativeQA and MultiFieldQA-en (Bai et al., 2024). Pool membership was originally defined by an OLMo input-length threshold and is retained after Llama retokenization. The Llama inputs range from 3694 to 16304 tokens, with 316 within 8192 tokens and 315 above it. Thus

the pool tests natural inputs through roughly twice the Llama native window, while the same installed table has scale $s = 4$. It is this frozen finite source pool, not the full LongBench benchmark.

The two complete arms use the same prompt hashes, references, order, decoder, per-task caps, batch 1 and unpadded prefill chunks of 8192. Their saved contracts differ only in arm label and static table. Both reuse the classic TailSpline/MrPro tables and common gain. Output caps are 32 for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. All 1262 completed answers were rescored with the recorded whole-response normalized token-F1 scorer; no empty or capped output was discarded.

There are 524 distinct source-context clusters, with no cluster shared across tasks. The primary estimand averages question F1 within each task and then gives the five task means equal weight. Within each task, the paired bootstrap resamples source-context clusters and divides the sampled score sums by sampled question counts; 20,000 draws use seed 20260914. The within-native and extended strata use seeds 20260916 and 20260915, respectively. The portable score-only pairs retain the cluster and input-length identifiers and reproduce all three intervals in Table 41.

Table 41: **TailSpline natural QA**. Task-equal F1 with source-context cluster intervals. Both arms use extension tables, including within the native window.

Input stratum	N	T (%)	P (%)	T-P (pp), 95% CI
All (primary)	631	41.08	40.88	+0.20 [-1.53, +1.89]
$\leq 8K$	316	46.43	45.73	+0.70 [-1.25, +2.81]
$> 8K$	315	37.43	38.37	-0.94 [-4.15, +1.95]

Table 42: Natural-QA631 task means and complete-output health.

Task	N	TailSpline (%)	MrPro (%)	Difference (pp)
hotpotqa	166	52.98	53.09	-0.10
2wikimqa	173	36.96	35.19	+1.76
qasper	119	40.73	41.04	-0.31
narrativeqa	61	22.82	23.77	-0.95
multifieldqa_en	112	51.91	51.33	+0.58
EOS	—	625	624	—
Cap-hit	—	6	7	—
Empty	—	0	0	—
Generated tokens	—	6972	7019	—

Overall T-P is +0.1957 points, with interval $[-1.5311, 1.8864]$. The extended stratum is -0.9383 points $[-4.1494, 1.9539]$, and the within-native stratum is $+0.6998$ $[-1.2478, 2.8073]$. The 315 extended questions have task counts 142/52/11/53/57 in the order listed above, drawn from 142/52/9/5/46 source contexts. Their smaller number of source contexts, particularly for NarrativeQA and Qasper, is retained in the clustered intervals. These are T/P extension-table contrasts, not comparisons against original Native.

For the document-equal sensitivity, question scores are first averaged within source context, then contexts within task, and tasks equally. T/P macro is 40.8141/40.8375%, a difference of -0.0234 points. This alternative weighting does not replace the frozen primary estimand. The observed natural-QA differences are much smaller than the clean RULER effect; their primary interval spans both signs. The result measures the natural-input behavior of TailSpline itself and is separate from the older BM/OLMo natural-QA result.

H.10 CLEAN CONFIRMATION AT THE INTERMEDIATE LENGTH

The 16K clean RULER panel evaluates the same $s = 4$ frozen constructions on 50 paired inputs per task, for 650 pairs across all 13 tasks. The primary score is the task-equal mean at this length. The completed matched report uses 20,000 paired bootstrap draws within task and length (seed 20260923). It gives TailSpline/MrPro 86.0949/82.7051% and difference $+3.3897$ points, with

interval [1.5307, 5.3385]. This is a separate length panel with its own inputs and sample count. Retrieval improves by 4.50 points, with interval [2.50, 6.69]; QA improves by 8.00 points, with interval [1.00, 15.00]. Aggregation and tracking contrasts are -2.77 and -2.40 points, respectively. The full task-equal aggregate retains these families. Complete task means follow.

Table 43: Clean RULER at 16K: 50 paired inputs per task. All task means are retained.

Task	TailSpline (%)	MrPro (%)	Difference (pp)
cwe	83.80	80.00	+3.80
fwe	25.33	34.67	-9.33
niah_multikey_1	100.00	88.00	+12.00
niah_multikey_2	98.00	86.00	+12.00
niah_multikey_3	92.00	84.00	+8.00
niah_multiquery	98.00	97.00	+1.00
niah_multivalue	98.50	97.50	+1.00
niah_single_1	100.00	100.00	+0.00
niah_single_2	100.00	98.00	+2.00
niah_single_3	100.00	100.00	+0.00
qa_1	76.00	70.00	+6.00
qa_2	60.00	50.00	+10.00
vt	87.60	90.00	-2.40

I FREQUENCY SPECTRA AND LEARNED ASSIGNMENTS ARE DISTINCT

A positional basis enters attention through learned Q/K projections. We test three separate changes: retargeting its range, swapping tables across trained weights, and reassigning the same frequencies to different slots. Each intervention addresses a different explanation of model behavior.

I.1 CHANGING THE EVALUATION RANGE CHANGES THE PREFERENCE

Return to the paired models of §3.1, without further training. The retained-range condition uses base $b = 256$ at every evaluation length. The retargeted condition replaces it with $b = L_{\text{eval}}$ for both allocations, keeping each trained weight set and normalized shape z fixed (the exact rule is in Appendix B.1). Cosh-minus-geometric NLL becomes $+0.060/+0.227/+0.460$ at 512/1024/2048 tokens: all three seeds now favor the geometric allocation (Fig. 16a). The same trained pair produces opposite orderings under the two range policies. Allocation and range are separately controllable, but their effects interact in the learned model.

I.2 CROSSING WEIGHTS WITH THEIR FREQUENCY TABLES

To isolate learned compatibility, hold each weight set fixed and cross the runtime tables. In a 50M pair, Geo and Cosh achieve matched PPLs of 7.14 and 7.16. Installing the Cosh table into Geo-trained weights raises PPL to 76.20; the reverse swap gives 23.05. A 151.9M crossing repeats the preference across two training seeds using factor-four tables derived from each training allocation (Fig. 16b). The two seed-specific interaction contrasts are 3.400 and 3.251 tail NLL. These crossed comparisons expose the learned association: both allocations support competent models, and each model learns to use its own basis (full crossings in Fig. 16).

I.3 FREQUENCY IDENTITY AND SLOT IDENTITY

The distinction becomes sharper when the frequency multiset itself is preserved. For a block permutation P corresponding to a frequency permutation π ,

$$\mathcal{R}_{\pi\Omega}(\Delta) = P\mathcal{R}_{\Omega}(\Delta)P^{\top}, \quad (Pq)^{\top}\mathcal{R}_{\pi\Omega}(\Delta)(Pk) = q^{\top}\mathcal{R}_{\Omega}(\Delta)k. \quad (68)$$

Thus permuting Q/K blocks together with frequencies exactly preserves the attention kernel. With weights fixed, however, permuting only internal frequency assignments raises OLMo tail NLL from

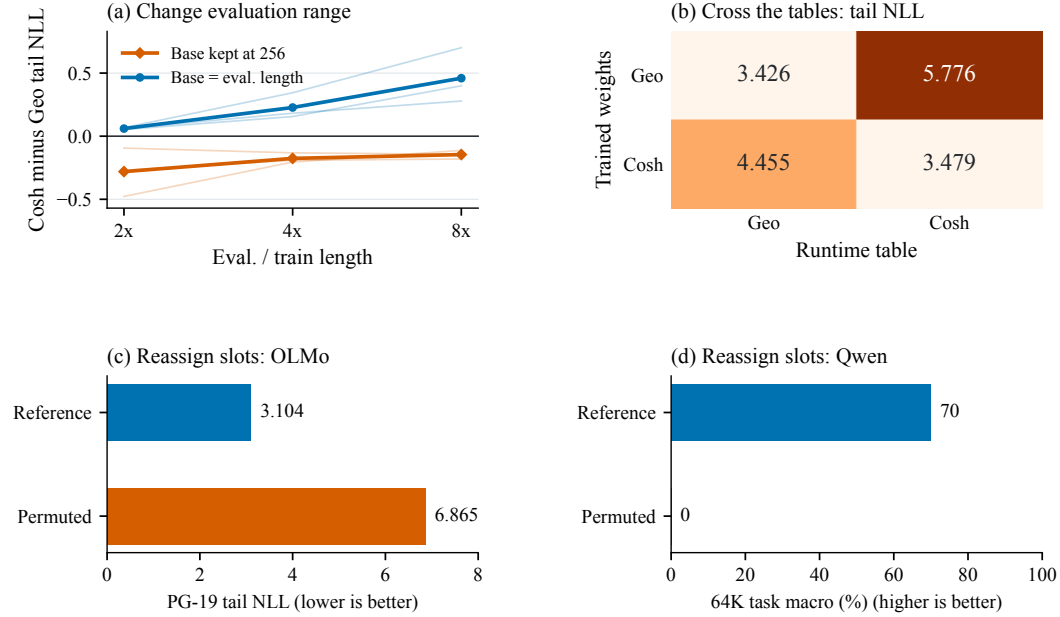


Figure 16: **Three interventions distinguish the supplied spectrum from learned compatibility.** (a) Range retargeting reverses the ordering of the same trained allocations; thin lines are three seeds and thick lines their means. (b) Crossed factor-four tables at 1K: rows are trained weights, columns runtime tables, and cells are mean tail NLL over two seeds and 32 anchors/seed. (c,d) Permuting interior frequency assignments preserves the final frequency multiset, endpoints and gain, but changes OLMo PG-19 tail NLL and Qwen 64K task macro (4 tasks, 20 rows/task). Each metric has its own axis. Joint Q/K compensation is the algebraic identity in (68), not a plotted recovery experiment.

3.10423 to 6.86493 and lowers Qwen 64K task macro from 0.7000 to 0.0000. The model uses a specific association between coordinate subspaces and frequencies (Fig. 16c,d).

For genuinely different frequency multisets, position-independent invertible Q/K maps cannot equate the rotation kernels over an interval containing zero (Theorem 6), or at integer distances 0, 1 when frequencies lie in $(0, \pi)$ (Corollary 2). Allocation changes the supplied basis; assignment changes its association with learned coordinates. The crossed-table controls show that an allocation learned during training need not be a drop-in replacement for a frozen checkpoint. These controls complement the constructive learning and deployment results in the main text.

I.4 ARCHIVED SLOT-INTERVENTION PROTOCOL

These historical comparisons permute the final frequencies of a static log-interpolated table, leaving slots 0 and 63 fixed. The OLMo reference uses $L_0 = 4096$, $b = 500,000$, $K = 64$, $s = 4$ and gain $1 + 0.074 \ln 4$; its table is the frozen log-law profile, not TailSpline. Its reported $1 \times$ PG-19 reference contains 20 documents and scores tail NLL. Qwen-2.5-1.5B recomputes that profile for $L_0 = 32768$, $b = 10^6$, retains $K = 64$, $s = 4$ and the same gain, and evaluates 20 rows per task at 64K for single-key retrieval, multikey-2, multikey-3 and variable tracking. The source report identifies the reference tables by SHA-256 prefixes 56ddfae2800d (OLMo) and ed8abbb27a1a (Qwen).

These interventions change the frequency-to-coordinate assignment while preserving the supplied spectrum. The trained weights are fixed. The complementary crossed-table experiment in Appendix B.2 changes the supplied spectrum and measures its compatibility with each trained checkpoint.

J FIXED-TABLE QUALITY OVER A DEPLOYMENT INTERVAL

J.1 THE DEPLOYMENT OBJECTIVE AND ITS DESIGN VARIABLES

Let L be a checkpoint’s native window and $H = SL$, $S > 1$, its declared deployment horizon. For a task-valid minimum length $\ell_{\min} > 0$, the objective concerns $[\ell_{\min}, H]$. The same table and amplitude are used at every length; neither changes with the current prompt length or during decoding. For a fixed checkpoint, write

$$\omega'_k = \omega_k^N S^{-m_k}, \quad m_k = cb_k, \quad 0 \leq b_0 \leq \dots \leq b_{K-1} = 1, \quad 0 \leq c \leq 1. \quad (69)$$

Within this monotone family, active increments of b locate the transition, their normalized distribution specifies its shape, and c controls the terminal compression. This is a search parameterization; it does not assert that an optimal unrestricted table must have one contiguous band. Changing c scales both the transition and the slow plateau. Comparisons of depth therefore measure whole-profile depth effects, rather than an isolated low-frequency intervention. Allowing the terminal depth to vary also changes sampled support, unlike the identification intervention in §3.1.

For higher-is-better task quality $Q_\theta(r)$ at length rL , one useful interval summary is

$$A_{[1,S]}(\theta) = \frac{1}{\log S} \int_1^S Q_\theta(r) d \log r. \quad (70)$$

This extended-window summary accompanies evaluations below L , native and endpoint values, the worst measured length, and task-family changes. Different tasks can have different valid short-input ranges. NLL and task accuracy retain their own units and aggregates. On a finite length grid, we use trapezoidal integration in log length. The measured minimum is not a certificate of the continuous interval minimum.

Baseline curves use a fixed table under the same evaluation contract. A pointwise envelope of several baselines is an oracle reference for regret, not a deployable method that may switch tables. For RULER we score the complete generated response with the official task scorer; normalized full-answer exactness, EOS, cap exhaustion, and natural-QA response F1 are separate measurements. This distinction matters when a model produces the requested answer and then continues generating.

J.2 A CONDITIONAL ALTERNATIVE TO FULL TERMINAL COMPRESSION

A slow plateau with frequency multiplier $a = S^{-c}$ maps separation rL to native effective separation arL . Full compression $a = 1/S$ exactly matches native phase exposure at the endpoint. It need not minimize distortion over the intervening lengths.

Proposition 12 (Endpoint and interval optima for a scalar slow-block objective). *Let $S > 1$, $a \in [1/S, 1]$, and let F be continuously differentiable on $[1/S - 1, S - 1]$. Define*

$$J(a) = \frac{1}{\log S} \int_1^S F(ar - 1) \frac{dr}{r}. \quad (71)$$

Then

$$J'(a) = \frac{F(aS - 1) - F(a - 1)}{a \log S}. \quad (72)$$

If $F(z) = f(|z|)$ with f strictly increasing, the unique minimizer is

$$a_* = \frac{2}{S + 1}, \quad c_* = \log_S \frac{S + 1}{2}. \quad (73)$$

The endpoint objective $F(aS - 1)$ is instead uniquely minimized at $a = 1/S$.

Proof. Differentiating under the integral and substituting $z = ar - 1$ gives $J'(a) = (\log S)^{-1} \int_1^S F'(ar - 1) dr$, hence (72). On the feasible interval, $a - 1 \leq 0 \leq aS - 1$. Strict increase of f makes the derivative negative when $aS - 1 < 1 - a$ and positive when the inequality reverses. Equality gives a_* and establishes uniqueness. At the endpoint, zero error is attained uniquely at $a = 1/S$. $\square$

This balances pull-in at the native length against overshoot at the horizon in a declared scalar objective. It does not identify the true task loss of a checkpoint. For $S = 2, 4, 8$, the conditional depths are 0.584963, 0.660964, and 0.723308, respectively. These values define concrete experimental controls against $c = 1$; they are not tuned task results.

Native and endpoint emphasis change the answer. For any probability measure over $r \in [1, S]$, a quadratic objective with native and endpoint weights $\lambda, \eta \geq 0$ has

$$J_{\lambda, \eta}(a) = \mathbb{E}[(ar - 1)^2] + \lambda(a - 1)^2 + \eta(aS - 1)^2, \quad (74)$$

$$a_* = \frac{\mathbb{E}[r] + \lambda + \eta S}{\mathbb{E}[r^2] + \lambda + \eta S^2}. \quad (75)$$

The second derivative is strictly positive, and $a_* \in [1/S, 1]$. Under log-uniform measure and $\lambda = \eta = 0$, this reduces to (73). Native emphasis moves the optimum toward 1; endpoint emphasis moves it toward $1/S$. Thus the chosen operating distribution and preservation priorities belong in the objective, rather than being determined by S alone. This scalar analysis concerns exposure relative to L ; real short-input task quality still requires direct measurement.

An asymmetric example is $F_\kappa(z) = \kappa z^2$ for $z \geq 0$ and z^2 otherwise, with $\kappa > 0$. Equation (72) yields

$$a_* = \frac{1 + \sqrt{\kappa}}{1 + S\sqrt{\kappa}}.$$

A larger overshoot penalty moves the optimum toward full compression. The formula does not specify a checkpoint's κ ; that quantity requires an independently justified measurement or loss model.

A circular counterexample. Even symmetry alone is insufficient. Set $F(z) = 1 - \cos(2\pi z)$ and $S = 4$. At $a = 2/5$, the two endpoint errors are $\pm 3/5$, so $J'(a) = 0$, but

$$J''(2/5) = \frac{25\pi \sin(6\pi/5)}{\log 4} < 0.$$

The balanced point is a strict local maximum. Consequently, neither periodicity nor successful endpoint phase matching implies the monotone distortion assumed by Proposition 12. The proposition supplies an interpretable comparison, while finite-phase and task measurements decide whether it helps a real model.

J.3 RECORDED DEVELOPMENT EVIDENCE FOR INTERVAL EVALUATION

Table 44 records two separate development panels. The Llama-3-8B-Instruct $S = 8$ comparison holds the C42V24 transition shape and gain fixed while remapping its band. The tasks are single-key (single-1), multi-key (multikey-1), and multi-value retrieval, with four prompts per task and length. Inference uses BF16 and greedy decoding, with chunked prefill and a full KV cache for long inputs. Both profiles reach 81.25% at 64K, but their 32K scores differ by 12.5 points. Endpoint evaluation alone misses this difference. Moving the band also changes total displacement; this is a band-remapped profile comparison, not a fixed-displacement intervention.

The OLMo-2-1.485B-Instruct $S = 4$ mini panel comprises six tasks, three lengths, and 18 prompts per cell: 324 responses per table. The tasks are single-key retrieval (single-2), multi-key retrieval (multikey-2), multi-query retrieval, variable tracking, frequent words, and QA-1. Cell means are averaged equally across tasks before computing AUC. The remapped C42V24 profile improves endpoint quality relative to BM, with a native cost. The AUC difference is +0.92 points, with task-stratified paired-row bootstrap interval $[-2.48, 4.36]$ (20,000 resamples). Task-AUC differences include -8.61 points on variable tracking and -1.39 on QA. This is a measured trade-off, not a confirmed improvement over BM across the interval.

These comparisons show that intermediate lengths and task families reveal effects hidden by endpoint scores. They motivate evaluating each installed profile over its intended deployment interval.

Table 44: **A fixed table’s endpoint and interval quality can differ.** Official task scores in percent; each row uses one table at every listed length. Llama is a small development screen (12 prompts/length); OLMo is a development mini panel (108 prompts/length). Within each panel the inputs, scorer, and decoder are shared. All remapped profiles use the registered C42V24 transition shape. AUC is integrated over the listed log-length grid, using unrounded cell means in the original summaries. These panels are distinct from the earlier 350-row and natural-QA studies.

Llama, fixed $S = 8$	8K	16K	32K	48K	64K	AUC
C42V24 [14, 32]	95.83	97.92	100.00	95.83	81.25	96.62
C42V24 [16, 34]	97.92	97.92	87.50	95.83	81.25	93.67
OLMo, fixed $S = 4$	4K	8K	16K	AUC		
BM	81.67	71.31	48.50	68.20		
MrRoPE-Pro	42.28	25.63	8.18	25.43		
C42V24 [14, 31]	79.65	71.84	53.15	69.12		

J.4 FREQUENCY INTERACTIONS IN ATTENTION

For fixed pre-rotation Q/K coefficients, a rotary score can be written

$$\ell(\Delta; m) = \text{Re} \sum_k C_k e^{i\Delta\omega_k^N S^{-m_k}}, \quad \frac{\partial \ell}{\partial m_k} = \log S \Delta\omega_k' \text{Im}(C_k e^{i\Delta\omega_k'}).$$

The sign depends on content and separation. For a full row of visible keys, let J be the logit Jacobian with respect to m , including the actual attention scale and rotary amplitude. At the reference distribution p , the local attention-KL curvature is

$$H = J^\top (\text{diag}(p) - pp^\top) J.$$

It couples frequencies through key competition even though each score is a sum over pairs. If $m = T\epsilon$ is the cumulative-increment parameterization, the corresponding curvature is $T^\top HT$. Diagonal per-slot scores do not retain these interactions.

This calculation identifies a structural source of interaction: softmax normalization couples frequency changes through all competing keys. It explains why per-frequency geometric changes can combine differently in attention. Here the content coefficients remain symbolic; the Cosh and BM constructions use declared geometry and design parameters to form their tables. The fixed-support and crossing experiments test the resulting model behavior under their respective protocols.

J.5 EARLIER PHASE GEOMETRY AND PROFILE DIAGNOSTICS

For $s > 1$, $\omega_j' = \omega_j s^{-m_j}$ maps separation Δ to native effective separation Δs^{-m_j} . At $\Delta = sL_{\text{train}}$, containment within the unwrapped training interval requires $m_j \geq 1$. This is a per-pair geometric condition: distinct m_j generally map to distinct native separations. The phase derivative at $m_j = 0$ is $-\Delta\omega_j \ln s$, while the finite drift at the native table is zero.

For twelve OLMo-1B profiles on 350 shared 16K RULER inputs, a count of slots whose test-distance turns lie in $[0.25, 16]$ separates three response groups. Appendix J.6 defines this exploratory count and its reconstruction provenance. It leaves within-group rankings unresolved; it is not used to certify interval quality.

J.6 RECONSTRUCTING THE TWELVE-PROFILE RESPONSE ANALYSIS

Statistic and evaluation unit. For OLMo-2-0425-1B-Instruct, $K = 64$, $b = 500,000$, native window 4096, and test distance $D = 16384$, define

$$T_j(D; m) = \frac{D}{2\pi} b^{-j/K} 4^{-m_j}, \quad N(m) = \sum_{j=0}^{63} \mathbf{1}\{0.25 \leq T_j(D; m) \leq 16\}. \quad (76)$$

This is a band-count statistic. The thresholds are exploratory choices examined on the same twelve profiles, not calibrated-readout boundaries. All arms use $g = 1 + 0.1 \ln 4 = 1.1386294$, 350 shared

row identities, seven equally sized RULER task groups, and 16K inputs. The score is the mean saved task score in $[0, 1]$, including partial credit; EOS is recorded separately. The 350 inputs estimate each profile’s task score, whereas the configuration-level analysis has twelve profiles. MrPro and BM use archived endpoint outputs; the other ten arms have separate saved outputs.

Table 45: **All twelve profiles used for exploratory response fitting.** Scores are recomputed from saved row-level outputs. SHA prefixes identify reconstructed little-endian float32 frequency tensors; these are reconstruction digests, not recovered runtime tensor attestations. All twelve profiles participate in the fit.

Profile	N	Score sum	Mean (%)	EOS / 350	SHA-256 prefix
MrPro	15	24.80	7.09	197	505d1fb2dce3
beta_b0p25	16	50.65	14.47	181	a8313cac3ee8
turns_a0p5b32	16	56.15	16.04	169	25721495d712
beta_b0p5	16	81.20	23.20	150	4b0b97569a3a
turns_a1b16	16	92.80	26.51	167	a24e3329fc49
knife_both	17	134.60	38.46	173	4c9d6fa7684b
knife_taper	17	137.80	39.37	146	a3b8138d159e
knife_leak	17	138.90	39.69	120	8f92ffecc376
MrProBM	17	145.85	41.67	119	0ed902cba69b
beta_b2	17	175.05	50.01	119	1188a7d3c34c
turns_a2b32	17	178.50	51.00	129	cbd8c451c0bb
turns_a1b64	17	188.45	53.84	130	7aedd80ef093

Fit and scope. Unweighted least squares of the logit-transformed mean score on N gives $\text{logit}(\bar{r}) = -20.45 + 1.19N$. Among 66 profile pairs, 39 have different counts and concordant scores, 0 have different counts and discordant scores, and 27 tie in count. The count therefore describes between-group ordering and does not resolve within-group differences. This is a descriptive fit to fractional scores, not a binomial success model or an independent predictive test. Threshold choices and coefficients were analyzed using these same profiles.

Reconstruction and provenance. The accompanying `figs/profile_diagnostic_inputs.json` records every m_j , full reconstruction digest, row-output digest, score sum, and fit coefficient. The common row-identity digest is `14736f284ac30c0c`. Matching row identities checks panel alignment in the saved outputs; the available records do not supply a new prompt-byte comparison for the archived endpoints. The analysis therefore reports the response of the recorded profiles.

Testing the interpretation beyond the fitted family. A subsequent matched continuous-text evaluation at 16K on 16 documents increased the count from $N = 17$ for BM to 22–26 by releasing slow-end interpolation. BM had NLL 2.8627; the five modified tables gave 2.8971, 2.8674, 2.8697, 2.8878, and 3.0372. Every modification increased the count while worsening this likelihood endpoint. These NLL measurements reject the claim that increasing the count guarantees lower natural-text loss. They address a different endpoint from the RULER ordering fit, whose independent same-metric validation remains separate.

The archived BM endpoint is also not an exact text-replay target: a later same-name rerun differed on 24/350 row scores and 210/350 generated texts despite tiny reconstructed tensor differences. Table 45 reports the saved archived outputs; it does not pool them with reruns.